SplatAtlas: What Photometric Rankings Leave Unresolved in Gaussian Splatting

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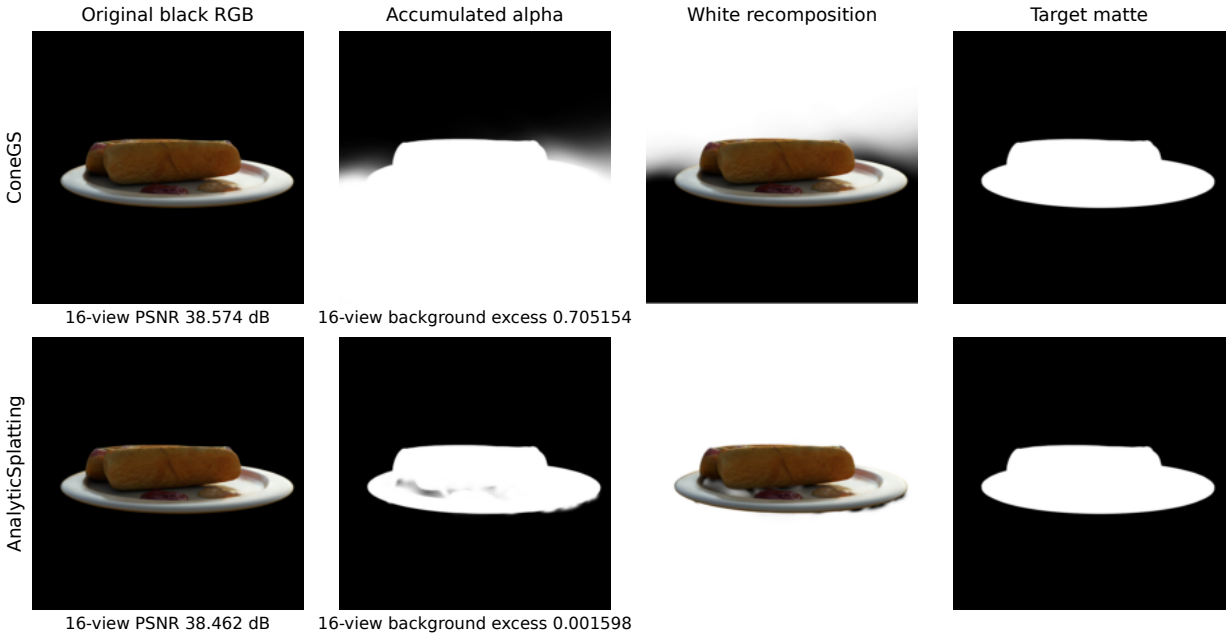


Figure 1. Hotdog, fixed camera 0: original RGB, accumulated alpha and white-background recomposition of the same retained states. GT matte is shared. State summaries shown below the panels use the sixteen task views and pooled-MSE PSNR; the eighty-view direct RGB comparison in Section 4.3 is a separate measurement. Both states belong to the Hotdog near-best quality set.

Abstract

Photometric rankings evaluate rendered images but do not establish the foreground-recomposition behavior of a retained Gaussian field. SplatAtlas connects image evaluations and native measurements to saved states, enabling inventory-level inspection and targeted tests on the same assets. Its active shared inventory contains 1,023 states from 33 methods on 31 scenes. We evaluate foreground coverage on 240 states from 30 supported methods across eight scenes. Coverage errors exhibit recurring method-associated structure with substantial scene dependence. Among 61 states within 0.6 dB of their scene's best black-background PSNR, ten states from five methods have mean excess background alpha of at least 0.05; five of these states belong to the extension beyond the initial discovery set. Fixed-state deletions further localize consequential differences in primitive roles. In one state, removing 49 primitives reduces mean excess background coverage by 0.3368 while changing original-background RGB by only 0.000186 LPIPS; comparable fractional selections in low-error reference states primarily affect correct foreground. Controlled training reverses the method ordering under fixed gray and nearly closes the coverage gap under random RGB. Attribute refinement sharply changes coverage on fixed geometric carriers, with state-dependent RGB costs. These results identify foreground coverage as learned attribute organization shaped jointly by background conditions and method design.

1. Introduction

A trained Gaussian field is both an image generator and a retained asset. Novel-view synthesis benchmarks compare the images it renders against reference views. Reusing the

field can ask for something else: an object rendered against a different background should preserve its foreground while exposing the replacement background around it. The original image score does not directly measure this decomposition. Foreground recomposition makes the distinction concrete through a task with a known object matte and a specified replacement background.

Photometric ambiguity establishes that matching RGB does not uniquely specify a foreground–background decomposition. The empirical question is how this freedom is realized in retained Gaussian states: how coverage errors are distributed across methods and scenes, which remain among high-scoring assets, and which primitive groups support them. A few contrasting outputs illustrate the possibility; a linked inventory makes its recurrence, magnitude and exceptions measurable. Returning to the checkpoints then allows the role of selected primitives to be tested within the assets that produced those outputs.

SplatAtlas provides this connection between comparisons and states. Its active shared inventory contains 33 methods on 31 scenes, with native image evaluations, field summaries and saved-state records. We evaluate a supported subset of 30 methods on all eight NeRF-Synthetic scenes using the same sixteen cameras per scene. An initial eight-method discovery set and a twenty-two-method extension remain separately identified. The full 240-state matrix retains low-scoring as well as high-scoring outputs, allowing the distribution of coverage errors and the behavior of near-best image-quality candidates to be read together.

The resulting picture has repeated structure and strong conditionality. Several methods recur in the high-background-error group across scenes, while large errors among near-best-PSNR states concentrate in Hotdog, Lego and Ship. Low-error candidates remain the majority in these quality sets. Fig. 1 connects one image-close Hotdog comparison to its alpha and recomposition outputs. Fixed-state checks reveal a further contrast: a small selected group can substantially reduce erroneous background coverage in one asset, while the corresponding fractional selection in a reference asset mostly changes correct foreground. Footprint controls and six additional states retain smaller effects and foreground costs alongside the strong examples. Background-controlled training and fixed-carrier refinement establish a distinct empirical finding: coverage ordering changes with the training background, and coverage can change substantially without moving the geometric carriers.

We make two contributions:

- **A retained-state measurement resource.** SplatAtlas links image evaluations, native measurements and task outputs to saved states, allowing inventory comparisons to be followed by additional measurements and interventions on the same assets.
- **An empirical study of foreground coverage and primitive roles.** We characterize method-associated, scene-dependent coverage errors across a supported inventory, identify their persistence among high-scoring assets, and use fixed-state comparisons to localize contrasting background and foreground roles. Background controls and fixed-carrier refinement further identify a training-conditioned attribute organization behind these coverage differences.

2. Related Work

Measuring Gaussian representations. Gaussian splatting optimizes explicit primitives for rendered-image reconstruction [8]. PSNR, SSIM and LPIPS measure comparisons with reference images [15, 18], while shared datasets establish common scene collections [2, 7, 10, 14]. Libraries and benchmark tools support rendering and comparisons of execution or representation costs [13, 16]. SplatAtlas connects these evaluations to saved fields, so a measurement can be followed by another task evaluation or a fixed-state inspection of the same object.

Color, transparency and foreground support. NGS studies false transparency and uses noise infill to expose and reduce opacity errors that static image evaluation can obscure [3]. AmbiSuR examines photometric ambiguity in primitive tails and blending for Gaussian-splatting surface reconstruction [11]. These studies provide context for examining what image agreement leaves unspecified. Our empirical object is the foreground-recomposition behavior of a retained inventory: its distribution across methods and scenes, the tail remaining under a fixed quality condition, and the spatial effects of selected primitive groups.

Retained assets and primitive organization. Anti-aliasing, densification and compact representations change the primitives left by different methods [1, 4–6, 9, 12, 17]. We keep each tested asset's native rendering path and inspect its saved organization. Scale- and footprint-based removals are measurement operations on these states. Their effects relate selected groups to the foreground and background content they support, with reference states and alternative selections preserving the comparison's practical range.

3. SplatAtlas and Measurement Protocol

3.1. Active inventory and linked states

SplatAtlas registers 34 methods on 31 shared scenes: twelve Tanks & Temples, nine Mip-NeRF 360, eight NeRF-Synthetic and two Deep Blending scenes. The registered grid contains 1,054 method–scene entries. After withdrawal of the GiGS checkpoints, the active shared inventory contains 1,023 states from 33 methods. Another 138 states on nine additional scenes bring the active collection to 1,161

Table 1. Resource and task support. Foreground and structural entries refer to retained inventory checkpoints. Variant and view counts describe evaluations, not independent training runs.

Collection	Methods	States / entries
Registered shared grid, 31 scenes	34	1,054
Active shared grid, 31 scenes	33	1,023
Additional records, 9 scenes	scene-specific	138
Registered foreground, 8 scenes	34	272
Valid foreground, 8 scenes	30	240
Structural panel, 4 scenes	6	12

states and 56,653 evaluated state-views. Tab. 1 separates the resource from the foreground-task and structural supports.

A state record links the saved checkpoint or export, stage, native renderer, camera list, target grid, output images and measurements. ConeGS/Hotdog, for example, identifies a 30,000-step checkpoint with 65,693 primitives. Its common-black color and alpha produce the sixteen-view foreground scores; the fixed-view illustration uses camera 0, and its 0.1% removal identifies 65 rows in that same primitive population. The link from state to outputs, task scores, selected rows and variant outputs lets a population observation be inspected at its source.

Native opacity, count and scale summaries supply storage context. Their population and coordinate conventions follow the method's exported representation; anchor-based Scaffold-GS and Octree-GS records retain their distinct storage meaning. Appendix A gives current native NVS records and field overviews. Native NVS evaluations preserve their recorded protocols, separately from the common-black foreground task below. In particular, native mean-view PSNR is not substituted for the task's pooled-MSE score.

3.2. Foreground task and image measurements

Let M be the ground-truth matte and $P = FM$ the premultiplied foreground formed from the source straight-RGBA image. We extract the native black-background color C after compositing and before final clipping, and normal accumulated alpha A. Prediction and target on background b are

$$I_b = C + (1 - A)b, \qquad G_b = P + (1 - M)b. \tag{1}$$

Premultiplication precedes resizing, and color and matte use the same filter on the native output grid. Black, gray and white composites reuse the same C, A, changing the compositing background while keeping illumination and the field fixed. The evaluated regions are background $\mathcal{B} = \{M < 0.01\}$, foreground $\mathcal{F} = \{M > 0.99\}$, and the intervening boundary. Background excess and foreground deficit are

$$B_{\text{err}} = \text{mean}_{p\in\mathcal{B}}(A(p) - M(p))_+, \qquad F_{\text{err}} = \text{mean}_{p\in\mathcal{F}}(M(p) - A(p))_+. \tag{2}$$

Regional errors average pixels in each nonempty frame region and then frames equally. Matte MAE averages $|A - M|$ over all pixels and frames. The primary task uses sixteen fixed views indexed $0, 5, \ldots, 75$. RGB PSNR first averages frame MSEs and then applies $-10 \log_{10}$. We retain raw and clamped-float outputs separately; the quality condition uses clamped-float black-background PSNR. Direct LPIPS compares the two specified rendered outputs, with its view support stated alongside the comparison.

3.3. Task supports and quality condition

The foreground registration covers 34 methods on eight scenes. Thirty methods support the fixed native color/alpha task on every scene, giving 240 valid states and 3,840 state-view evaluations. The other 32 registrations record withdrawn checkpoints, a predeclared task readout exclusion, or unsuccessful native-interface export. They remain separate from the measured task matrix, with their specific reasons in Appendix A.

The discovery set contains 3DGS, AnalyticSplatting, Mip-Splatting, LightGaussian, ConeGS, MiniSplatting, 3DGS-MCMC and TrimGS, selected after exploratory inspection. The extension adds the 176 supported states of the other twenty-two methods. Both groups retain all eight scenes. For each scene, we define the near-best quality set using only the common-black task score $p_{m,s}$:

$$Q_s = \{m : p_{m,s} \geq \max_{j\in V_s} p_{j,s} - 0.6 \text{ dB}\}, \tag{3}$$

where V_s contains the thirty valid task methods. This fixed same-target score condition is distinct from direct render-to-render comparison. Distribution summaries describe the evaluated states; query, pair and view counts describe their coverage. The twelve-checkpoint structural panel and its preselected groups are defined in Section 5.

4. Foreground Coverage Across the Retained Inventory

4.1. Recurring structure, heterogeneous magnitude

Fig. 2 displays the complete 30-by-8 matrix of background excess. Across its 240 states, 46 states from nine methods have $B_{\text{err}} \geq 0.05$, and 41 states from eight methods have $B_{\text{err}} \geq 0.10$. These fixed cutoffs describe mean excess alpha on ground-truth background. They count states at specified error magnitudes, rather than percentages of failed pixels.

Method-associated differences recur across scenes. Ranking methods by decreasing background excess, ConeGS, MiniSplatting, GHAP and CoAdaptGS occupy

the highest-error eight of thirty in all eight scenes. Octree-GS does so in six scenes and Scaffold-GS in four. At the other end, the largest eight-scene errors for AnalyticSplatting and Mip-Splatting are 0.001598 and 0.003298. The full matrix preserves magnitude as well as ordering: a repeated relative position can coexist with markedly different absolute errors across scenes.

Across the twenty-eight pairs of scene rankings, Spearman correlation has median 0.580 and range 0.331–0.932; Kendall's coefficient of concordance is 0.655. These describe cross-scene recurrence among the retained states. The raw matrix's descriptive decomposition and complete method summaries are in Appendix B. High-error recurrence and high-quality behavior answer different questions: GHAP and CoAdaptGS enter none of the near-best quality sets. We therefore examine those sets directly.

4.2. High image quality leaves a tail of large coverage errors

The eight Q_s sets contain 61 states from sixteen methods. Ten states from five methods have background excess at least 0.05, including nine states at or above 0.10. Fig. 3 shows all 61 observations, their discovery/extension identity and each scene's median. The large-error states occur in Hotdog, Lego and Ship; the other five scenes have no quality-set state above 0.05.

The upper tail and the distribution's low-error body coexist. Hotdog contains fourteen quality-set states, four above 0.10, while its median is 0.003414. Ship contains fifteen states with four above 0.10 and median 0.000938. Lego has six states, two at or above 0.05 and one at or above 0.10. Thus a scene can retain several consequential high-quality coverage errors while most of its near-best candidates have much smaller background excess.

The phenomenon also appears beyond the discovery methods. The extension contributes 44 quality-set states from eleven methods, five of which exceed 0.05 across three methods. On Hotdog, Octree-GS and Scaffold-GS obtain 38.385 and 38.437 dB black-background PSNR with background excess 0.431185 and 0.377574. Both are in Q_s. The extension's lower overall median error, 0.001065 compared with 0.004257 in the discovery set, accompanies these additional tail cases. The two groups were assembled for different roles, so their distributions establish the extent and exceptions within this inventory rather than population prevalence.

4.3. Similar rendered images, different recomposition

Fig. 1 returns to the same Hotdog camera for AnalyticSplatting and ConeGS. Their retained original-background outputs have direct LPIPS 0.003161 averaged over eighty shared views. This direct-image comparison is recorded separately from the sixteen-view quality strata. On the common sixteen-view task, their black-background scores are 38.462 and 38.574 dB, while background excess is 0.001598 and 0.705154. White-background PSNR is 27.867 and 3.516 dB. Both states belong to the near-best quality set, and the actual RGB outputs provide an additional comparison beyond that membership.

The alpha images locate the task difference. In ConeGS/Hotdog, 60.70% of ground-truth background has accumulated alpha at least 0.9, while the mean absolute composited color there is only 1.24×10^{-5}. Scaffold-GS/Hotdog shows the same type of combination over 30.07% of background, with mean absolute color 1.55×10^{-5}. These effective composited colors resemble the original black background and obscure the excess coverage in ordinary RGB. Background replacement exposes it. We next inspect which primitive groups affect this coverage in the retained checkpoints.

5. Spatial Roles of Primitive Groups

5.1. Fixed-checkpoint comparisons

The structural study first examines AnalyticSplatting, ConeGS and MiniSplatting on Hotdog and Drums. Scale selection ranks the maximum native activated axis and removes $k = \max(1, \lfloor qN \rfloor)$ primitives at fractions 0.01%, 0.1% and 1%. Footprint selection ranks native projected support averaged over scoring cameras 2, 22, 42 and 62. Its support is the sum of single-primitive pixel areas under the native filter and opacity threshold, clipped to the image window; selection uses neither ground truth nor accumulated transmittance. Count-matched random selections retain three fixed seeds per fraction.

The extension examines PixelGS, ReactGS and GaussianPro on Lego and Mic at 0.1%, with scale, footprint and all three random selections. These methods and scenes were frozen using the original RGB selection rule before inspecting their task errors. All twelve checkpoints keep their geometry, native rendering components and remaining attributes after deletion, without finetuning. The 124 variant groups produce 1,984 fixed-camera comparisons at the same sixteen evaluation views. Each comparison measures a selected group's joint effect within its own checkpoint.

5.2. Background and foreground roles in the same scene

Hotdog connects a high-quality task comparison to the structural operation. Removing the largest 0.1% in ConeGS removes 65 primitives, reduces background excess by 0.149728 and changes original-background RGB by approximately 0.000019 LPIPS. The corresponding selection in AnalyticSplatting removes 172 primitives and reduces background excess by only 0.000010. Instead, 99.65% of

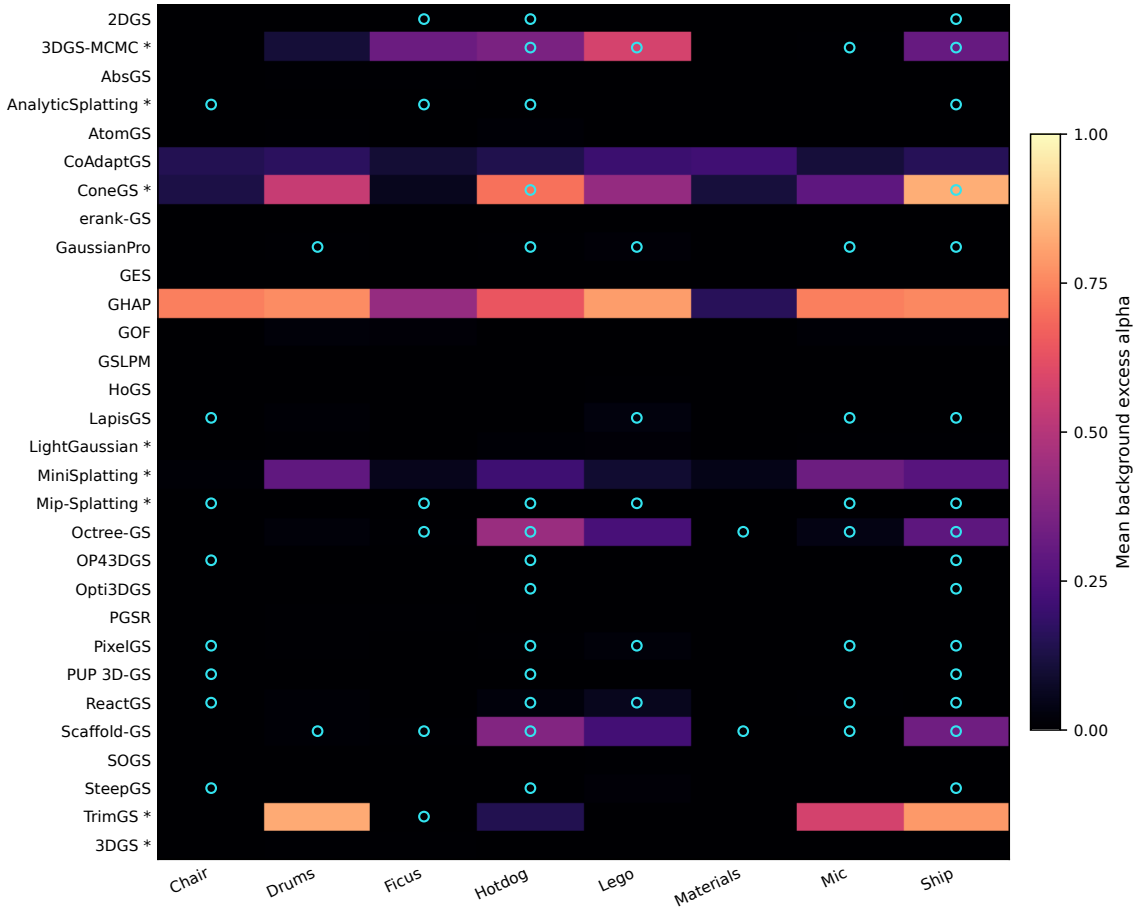


Figure 2. Complete valid foreground inventory: thirty methods and eight scenes. Every cell is the mean background excess over sixteen fixed cameras. Rows follow method ID; an asterisk identifies a discovery method and a cyan circle identifies frozen RGB-only quality-set membership. The common 0–1 color scale preserves absolute magnitude across all rows and scenes.

Table 2. All twelve checkpoints at 0.1% deletion. S/F pairs report scale and footprint selections, respectively. Positive ΔB reduces background excess; positive ΔF increases foreground deficit. Direct LPIPS compares the original-background output before and after deletion over the same sixteen views. Row numbers match Figure 4.

State	k	ΔB S / F	ΔF S / F	Direct LPIPS S / F
1. ConeGS/Hotdog	65	0.149728 / 0.150912	0.000195 / 0.000332	0.000019 / 0.000008
2. AnalyticSplatting/Hotdog	172	0.000010 / 0.000014	0.006011 / 0.011451	0.000499 / 0.001886
3. ConeGS/Drums	49	0.336849 / 0.406517	0.003231 / 0.004642	0.000186 / 0.000224
4. AnalyticSplatting/Drums	391	0.000218 / 0.000029	0.002530 / 0.006931	0.003090 / 0.006216
5. MiniSplatting/Hotdog	107	0.013613 / 0.023949	0.000142 / 0.000129	0.000168 / 0.000123
6. MiniSplatting/Drums	104	0.027816 / 0.038656	0.000085 / 0.000140	0.000160 / 0.000114
7. PixelGS/Lego	980	0.001817 / 0.003523	0.001376 / 0.002084	0.000333 / 0.001031
8. PixelGS/Mic	456	0.000065 / 0.000109	0.000175 / 0.000240	0.000093 / 0.000147
9. ReactGS/Lego	491	0.001742 / 0.012794	0.000206 / 0.000872	0.000053 / 0.000421
10. ReactGS/Mic	239	0.000851 / 0.002092	0.000070 / 0.000273	0.000012 / 0.000084
11. GaussianPro/Lego	1003	0.000630 / 0.001373	0.000898 / 0.001611	0.000305 / 0.000940
12. GaussianPro/Mic	1132	0.000301 / 0.000234	0.000112 / 0.000174	0.000082 / 0.000185

its absolute alpha-change mass lies in ground-truth foreground; foreground deficit increases by 0.006011 and matte MAE worsens by 0.001407. Fig. 4 makes these spatial roles visible on the same camera.

Tab. 2 gives both selectors across all twelve states. ConeGS/Drums supplies a stronger local background ef-

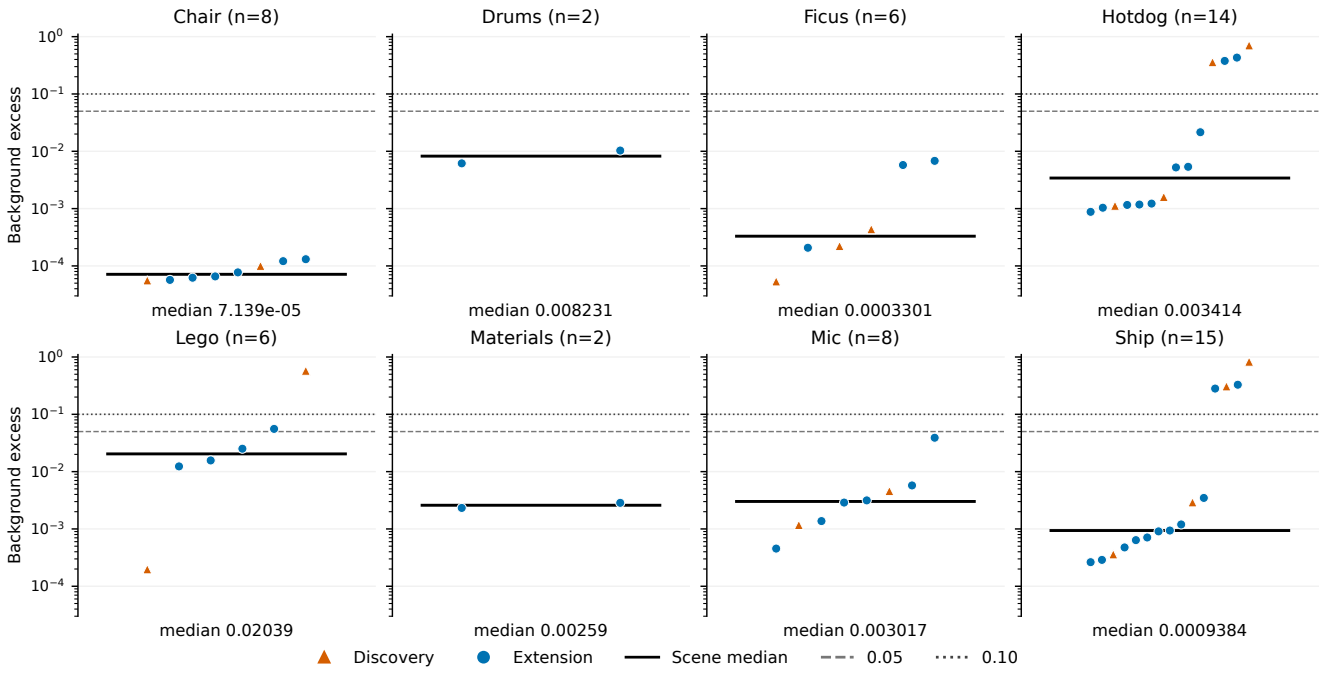


Figure 3. Every state in the eight near-best quality sets: sixty-one points, colored by discovery/extension role. The vertical scale is logarithmic; black segments mark scene medians and reference lines mark the predefined 0.05 and 0.10 descriptive cutoffs. Scene counts and medians retain the low-error body alongside the tail.

fect: deleting 49 scale-tail primitives reduces background excess by 0.336849 and matte MAE by 0.260579. Its direct baseline-to-variant pooled RGB PSNR is 49.8255 dB, with LPIPS 0.0001858; 98.14% of absolute alpha-change mass lies in ground-truth background. This checkpoint illustrates a large within-state effect and is outside the Drums near-best quality set. The Hotdog comparison supplies the explicit connection to a high-quality member.

The spatial shares use absolute alpha-change mass within each ground-truth region, normalized by the whole-image change and then averaged over views with a positive denominator. They locate where the selected group's removal changes coverage. The before/after differences are the overall effect of removing the group together; compositing does not turn them into an additive decomposition of independent primitive contributions.

5.3. Footprint controls and extended states

Footprint selection reduces more background excess than scale selection in ten of twelve states. At the same primitive count it also selects greater total projected support. The selectors' 0.1% sets have Jaccard overlap ranging from 0.059 to 0.607. Equal fractional removal therefore corresponds to different counts across states and different support areas within a state.

The fixed area-matching comparison finds common support only in AnalyticSplatting/Hotdog among the six original states. Its common target contains 163 primitives, retaining 97.60% of positive-support scale-selected primitives and 87.17% of their area. The three evaluated control-to-target area ratios are 0.9367, 0.9316 and 0.9306. The other five states do not meet the common-support requirements. These observations leave projected area as a viable explanation of differences between selectors and do not identify a separate three-dimensional-scale effect.

The six extended states retain smaller changes and visible costs. ReactGS/Lego's footprint selection reduces background excess by 0.012794, or 22.98% of its baseline, while increasing foreground deficit by 0.000872 and losing 0.24653 dB of black-background GT PSNR. The PixelGS/Lego and GaussianPro/Lego selections lose 0.60154 and 0.49451 dB and increase foreground deficit by 0.002084 and 0.001611. Their Mic states start with smaller background errors and retain small absolute improvements. Appendix D gives every fraction, random draw and matched-area comparison. Together these results establish state-specific group roles, including weak effects and damage to useful foreground content.

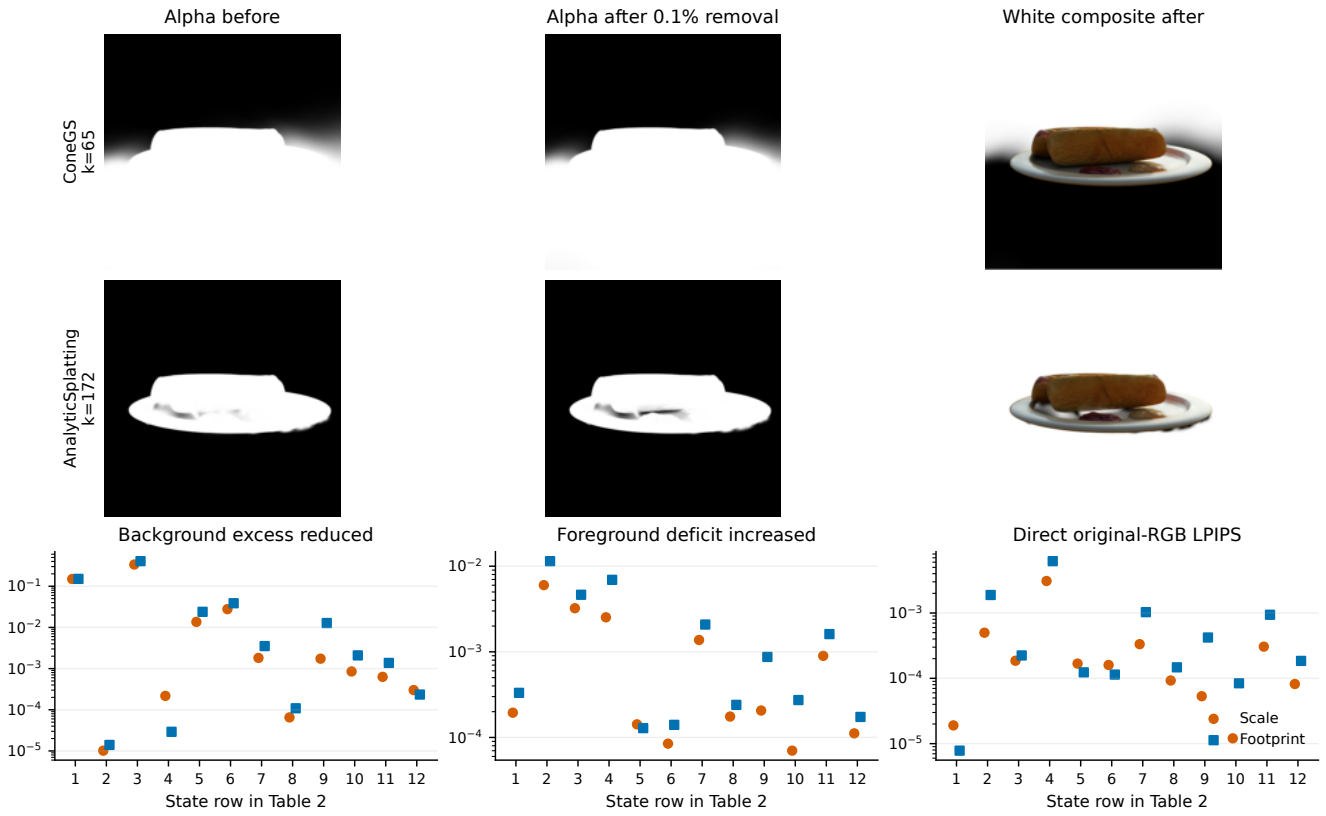


Figure 4. Fixed Hotdog camera 0 under the 0.1% native-scale selection. Alpha and white recomposition show background removal in ConeGS and changes to correct foreground in AnalyticSplatting. Lower panels retain all twelve state rows from Table 2 and compare scale and footprint selections at the same count within each state. Regional changes and direct RGB distances average sixteen cameras; lower-panel scales are logarithmic.

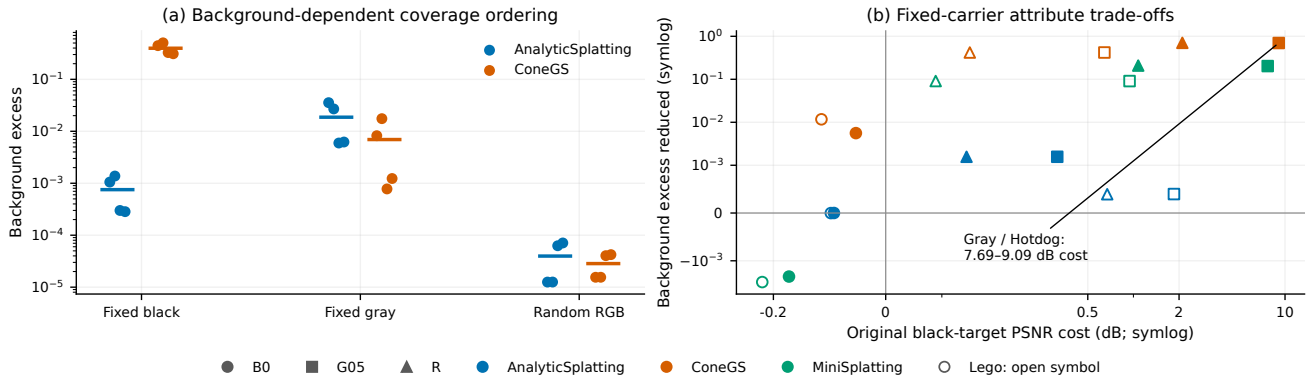


Figure 5. Coverage under controlled backgrounds. (a) All twenty-four core endpoints; segments mark four-setting means. Gray reverses the black-background ordering; random RGB reduces both errors. (b) All eighteen fixed-carrier branches versus black-target PSNR cost. Symlog axes retain negative changes and large costs; open symbols denote Lego. Endpoints use sixteen cameras.

6. Background-Conditioned Coverage Organization

6.1. Shared background controls

B0 uses black, G05 fixed RGB (0.5,0.5,0.5), and R independently samples each channel from Uniform[0,1] per update.

Table 3. Background-controlled training. Each row averages four scene–seed settings at 30k updates. Black and gray columns are PSNR (dB) against the corresponding targets.

Method	Bg.	B_{err}	F_{err}	Black	Gray
AnalyticSplatting	B0	0.000753	0.012924	38.33	32.85
AnalyticSplatting	G05	0.018672	0.002595	26.05	38.36
AnalyticSplatting	R	3.97e-05	0.000675	38.08	38.38
ConeGS	B0	0.396794	0.001492	38.66	13.03
ConeGS	G05	0.006937	0.009022	30.25	38.89
ConeGS	R	2.84e-05	0.000931	38.33	38.77

Training RGBA is composited against the same background as the prediction, using full-pixel RGB supervision without an alpha loss.

From-scratch training compares AnalyticSplatting and ConeGS on Hotdog/Lego with seeds 0 and 1: twenty-four 30,000-update endpoints. Three-background branches share initialization within each setting. Images, cameras and grid are common; native recipes are preserved and background RNG is separate.

6.2. Method ordering depends on the background

Under B0, all four paired settings retain close RGB but different coverage. AnalyticSplatting background excess is 0.000284–0.001377, compared with 0.310683–0.503786 for ConeGS; the mean ConeGS-minus-AnalyticSplatting gap is 0.39604. Direct black-background RGB comparisons give 42.820–45.509 dB and LPIPS 0.002910–0.003433.

G05 reverses the coverage ordering in all four settings: AnalyticSplatting background excess rises and ConeGS excess falls, giving a mean signed gap of -0.01174. The change also alters image behavior. Mean black-target PSNR falls to 26.05 and 30.25 dB, while gray-target PSNR reaches 38.36 and 38.89 dB, respectively (Tab. 3). A fixed color therefore changes the result before introducing cross-update variation.

R reduces background excess to 1.25×10^{-5} to 7.09×10^{-5} across all eight endpoints, leaving mean absolute between-method gap 1.42×10^{-5}. Relative to B0, black-target PSNR costs span 0.061–0.506 dB.

6.3. Attribute changes on fixed geometric carriers

We refine the six original AnalyticSplatting, ConeGS and MiniSplatting assets on Hotdog and Lego under the same backgrounds for 3,000 updates. Only opacity and SH appearance change; geometry, population, SH degree, cameras and applicable filters stay fixed. All eighteen geometry-preservation checks pass.

Under R, ConeGS background excess changes from 0.705154 to 0.000915 on Hotdog and 0.419939 to 0.000168 on Lego. MiniSplatting changes from 0.208784 to 0.000421 and 0.091135 to 0.000053. The black-target PSNR costs are 2.102 and 1.076 dB on Hotdog, compared with 0.150 and 0.089 dB on Lego. Foreground deficits increase by 0.000207–0.001714 across these four states. The low-error AnalyticSplatting references retain smaller absolute background gains and improved foreground coverage. Fig. 5 preserves all backgrounds, including gray branches with larger image costs.

Together, these studies establish coverage as training-conditioned attribute organization. The differences appear in rendered support, beyond native parameter conventions, yet selected fixed carriers support substantially different coverage after attribute refinement. In the tested methods and scenes, coverage is jointly shaped by the training-background regime and method design, with geometry admitting alternatives whose RGB and foreground costs vary.

7. Discussion and Conclusion

SplatAtlas connects an empirical distribution to inspectable retained assets. Coverage errors recur across methods and scenes, yet their magnitude and presence among high-quality candidates are conditional. The near-best sets contain a predominantly low-error body together with a multi-method tail on three scenes. Returning to the same checkpoints reveals sparse groups that materially affect background excess with little original-RGB change, alongside reference groups that mainly support correct foreground. The distribution and the group effects make foreground-recomposition behavior measurable beyond an image ranking.

Changing black to gray reverses method ordering, while random RGB nearly closes the coverage gap. Attribute-only refinement changes coverage on fixed carriers. At similar RGB, these differences reflect learned organization shaped jointly by background regime and method design, with state-dependent costs.

The foreground task evaluates replacement of the compositing background for the same RGBA object. Native NVS and foreground recomposition retain their respective targets and protocols. Method-associated recurrence describes a finite retained inventory, while deletion establishes local effects inside selected checkpoints. Count-matched and available area-matched controls preserve alternative spatial explanations; neither ranking consistency nor deletion identifies the historical training component that formed the fields. These boundaries situate the observations within the asset use actually tested.

Image metrics continue to measure rendered-image quality. Using a retained field for foreground recomposition calls for an evaluation of its foreground support as well. SplatAtlas makes those task measurements traceable to the saved states and primitive groups behind the images, enabling further comparison, inspection and reuse of the same assets.

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A. Inventory and Native Measurements

A.1. Image quality on the complete common-scene set

The table uses the native RGB outputs of the evaluated saved checkpoints on the fixed reference grids. NeRF-Synthetic uses black-composited reference images.

Table 4. PSNR (dB) on the common 31-scene set. All 30 methods have scores for every designated scene. Each entry is the equal-weight mean of scene scores, and each scene score averages its complete held-out RGB view set. Higher is better.

Method	NeRF-Synthetic 8 scenes	Mip-NeRF 360 9 scenes	Tanks and Temples 12 scenes	Deep Blending 2 scenes
3DGS	35.309	27.559	23.544	27.311
2DGS	35.145	27.379	22.990	27.086
3DGS-DR	34.572	26.588	21.787	25.981
3DGS-MCMC[a]	35.150	27.899	23.646	27.129
AbsGS	35.168	27.275	23.076	27.070
AnalyticSplatting	35.167	27.563	23.382	27.112
AtomGS	33.689	27.409	23.520	26.989
ConeGS	34.714	27.690	22.490	27.803
erank-GS	34.692	26.397	21.077	25.766
GaussianPro	34.876	27.456	23.270	26.866
GES	34.606	27.015	23.232	27.275
GHAP[f]	33.371	26.506	23.201	27.389
GOF	34.802	27.178	22.735	27.018
GSLPM	26.271	21.885	20.036	25.026
HoGS	34.720	27.123	22.776	26.393
LapisGS	35.276	27.051	22.805	24.313
LightGaussian[b]	33.432	27.032	22.209	25.121
MiniSplatting	34.475	27.304	22.665	27.553
Mip-Splatting[c]	35.599	27.660	22.742	26.256
Octree-GS[e]	35.348	27.534	23.805	27.859
OP43DGS	35.155	27.448	23.079	27.040
Opti3DGS	35.111	27.397	23.385	27.408
PGSR	33.563	26.891	22.431	27.058
PixelGS	35.488	27.593	23.270	26.367
PUP 3D-GS	35.177	27.511	23.267	26.970
RaDe-GS[d]	33.783	27.097	22.437	27.589
ReactGS	35.376	27.515	22.802	25.996
Scaffold-GS	35.233	27.704	23.801	27.966
SOGS	34.669	27.517	23.278	27.512
SteepGS	35.152	27.409	23.249	27.037

[a]3DGS-MCMC uses the completed SfM-initialization repairs for auditorium, lighthouse, palace and temple; the remaining scenes retain their evaluated configurations. [b]LightGaussian uses 10k checkpoints for stump, playground and truck. [c]Mip-Splatting uses a 20k checkpoint for ballroom. [d]RaDe-GS uses the cap-5M repairs for bicycle, stump and treehill. [e]Octree-GS uses 40k checkpoints. [f]GHAP retains GMR at 15k with ratio 0.1. The other saved checkpoints are at 30k. These scores summarize the evaluated models and configurations.

A.2. Native storage context and image conditions

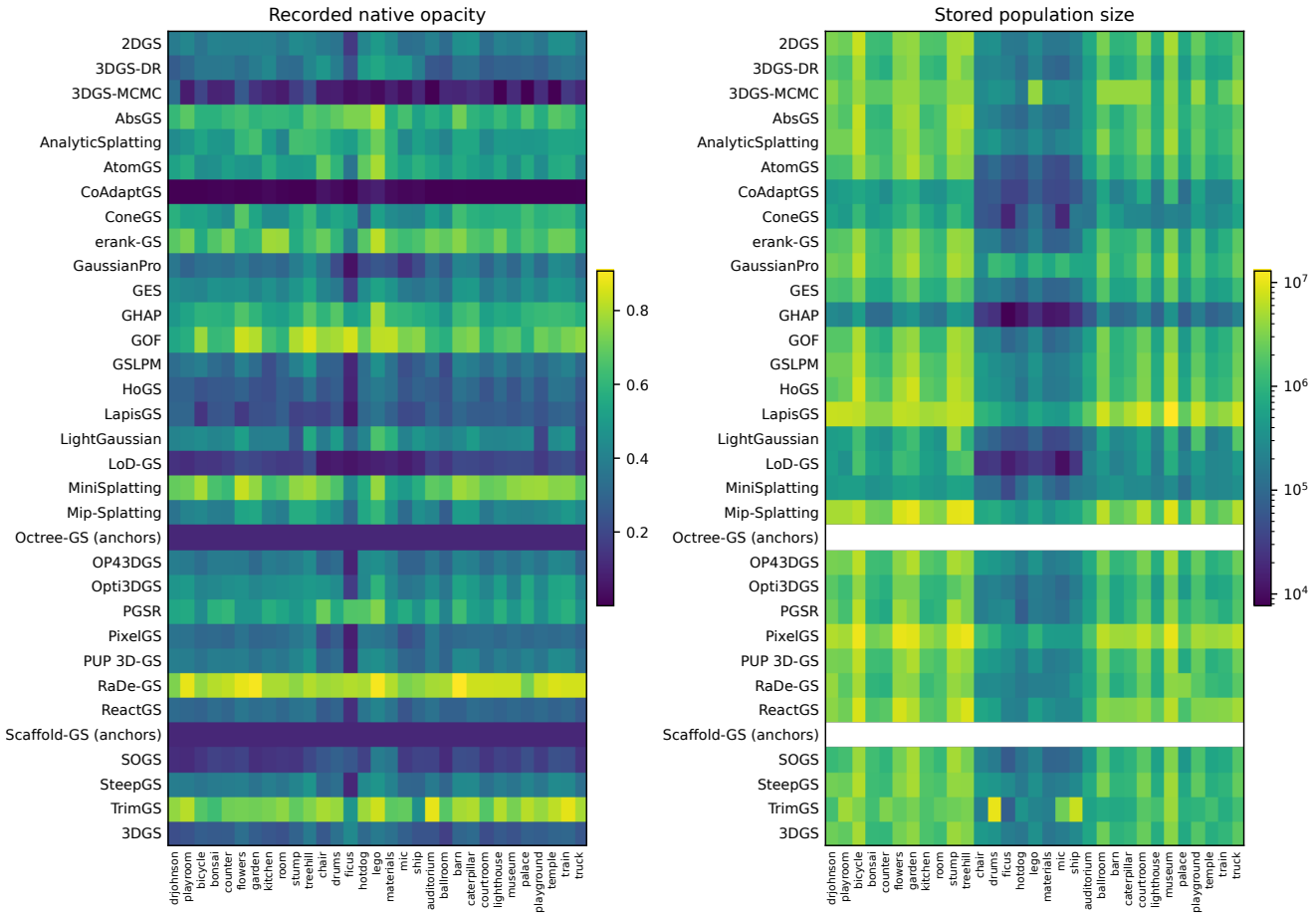


Figure 6. Native opacity records and stored-population sizes on the corrected active shared inventory. Anchor populations are identified in the row labels; their recorded opacity retains the anchor storage convention. These are storage summaries, separately from the foreground task.

Table 5. Current shared-scene image-conditioned native-opacity summaries. Pair counts refer to the full image pool; ordinary-primitive opacity excludes the two anchor populations. Scene-equal summaries first take a within-scene pair median. Method-balanced summaries first take each method's median incident-pair gap, then methods and scenes. PSNR labels give the gap upper bound; RGB percentages select the lowest direct-LPIPS fraction within each scene's pool, and LPIPS labels give a fixed direct-distance upper bound.

Recorded image condition	Pairs	Scenes	Pair-first gap	Method-balanced gap
All pairs	16368	31	0.192635	0.159886
PSNR 0.3 dB	4192	31	0.146561	0.148854
PSNR 0.3 dB; RGB 50%	2104	31	0.110370	0.108651
PSNR 0.3 dB; RGB 25%	1059	31	0.081852	0.076632
PSNR 0.3 dB; RGB 10%	434	31	0.065401	0.057899
PSNR 0.6 dB	7204	31	0.156244	0.148940
PSNR 0.6 dB; RGB 50%	3609	31	0.123710	0.106265
PSNR 0.6 dB; RGB 25%	1811	31	0.090423	0.085092
PSNR 0.6 dB; RGB 10%	736	31	0.074346	0.059611
PSNR 0.6 dB; LPIPS 0.015	1286	9	0.144818	0.138041
PSNR 0.6 dB; LPIPS 0.02	1441	10	0.147859	0.135321

Recorded image condition	Pairs	Scenes	Pair-first gap	Method-balanced gap
PSNR 0.6 dB; LPIPS 0.025	1612	12	0.147757	0.125579
PSNR 0.6 dB; LPIPS 0.03	1788	14	0.142304	0.122856
PSNR 1 dB	9539	31	0.166919	0.149343
PSNR 1 dB; RGB 50%	4778	31	0.138051	0.119794
PSNR 1 dB; RGB 25%	2396	31	0.097958	0.101697
PSNR 1 dB; RGB 10%	967	31	0.076969	0.070473

A.3. Foreground registrations and state links

The foreground registration includes 272 method–scene entries. GiGS is withdrawn on all eight scenes; RaDe-GS retains the predeclared foreground-readout exclusion on all eight. The 3DGS-DR foreground interface does not supply the required final color/alpha decomposition with its environment reflection; the LoD-GS foreground export fails its adapter path. Those two eight-state sets contribute no task scores. The valid thirty-method matrix is complete and does not replace these thirty-two registrations with numerical errors.

The ConeGS/Hotdog state is identified by `conegs_Hotdog/source/point_cloud/iteration_30000/point_cloud.ply`. Its 65,693 primitive rows, original camera IDs, native color/alpha outputs, sixteen-view task records and selected-row files remain associated with the same checkpoint. Evaluation cameras are 0,5,...,75. Footprint scoring uses 2,22,42,62. The direct Hotdog method-pair record uses eighty shared cameras; it is not the sixteen-view quality score.

B. Complete Foreground Results and Inventory Structure

Table 6. All 240 valid foreground states. Every row uses the same sixteen scene cameras. RGB scores are clamped-float pooled-MSE PSNR; regional and matte errors average frames equally.

Scene	Method	Black PSNR	White PSNR	Background excess	Foreground deficit	Matte MAE
Chair	2DGS	35.984	35.900	0.000079	0.000699	0.000664
Chair	3DGS-MCMC	36.316	34.906	0.000114	0.005990	0.001899
Chair	AbsGS	26.951	32.931	0.001313	0.000443	0.004596
Chair	AnalyticSplatting	36.443	36.056	0.000056	0.001184	0.000721
Chair	AtomGS	33.502	33.331	0.000161	0.000855	0.000936
Chair	CoAdaptGS	24.566	15.900	0.148275	0.021314	0.124724
Chair	ConeGS	35.682	13.190	0.125141	0.002941	0.103093
Chair	erank-GS	35.857	35.825	0.000060	0.000596	0.000711
Chair	GaussianPro	31.381	37.849	0.000764	0.000901	0.001740
Chair	GES	35.862	35.669	0.000059	0.000761	0.000721
Chair	GHAP	2.292	26.356	0.732695	0.000329	0.588013
Chair	GOF	36.055	35.853	0.000059	0.001227	0.000703
Chair	GSLPM	26.876	26.890	0.000068	0.000567	0.000678
Chair	HoGS	35.849	35.736	0.000057	0.000619	0.000702
Chair	LapisGS	36.556	35.723	0.000131	0.000563	0.000749
Chair	LightGaussian	34.703	34.335	0.000341	0.000476	0.000975
Chair	MiniSplatting	35.465	23.180	0.010930	0.000842	0.009638
Chair	Mip-Splatting	36.919	36.400	0.000100	0.000699	0.000634
Chair	Octree-GS	36.169	32.580	0.000717	0.010412	0.003325
Chair	OP43DGS	36.440	36.128	0.000062	0.000654	0.000693
Chair	Opti3DGS	36.221	35.877	0.000082	0.000658	0.000726
Chair	PGSR	26.860	32.425	0.001445	0.001606	0.004885
Chair	PixelGS	36.854	36.324	0.000077	0.000569	0.000681
Chair	PUP 3D-GS	36.490	36.182	0.000057	0.000593	0.000670
Chair	ReactGS	36.650	36.090	0.000121	0.000681	0.000747
Chair	Scaffold-GS	35.798	32.953	0.000516	0.008082	0.002592
Chair	SOGS	36.129	34.740	0.000390	0.002130	0.001318
Chair	SteepGS	36.471	36.158	0.000066	0.000621	0.000684
Chair	TrimGS	35.798	35.802	0.000056	0.000900	0.000719
Chair	3DGS	26.921	33.199	0.001273	0.000704	0.004566
Drums	2DGS	27.040	25.968	0.000723	0.004096	0.003606
Drums	3DGS-MCMC	27.005	13.234	0.103821	0.012981	0.086938
Drums	AbsGS	23.754	23.180	0.004143	0.002009	0.014185
Drums	AnalyticSplatting	26.993	25.396	0.001167	0.007864	0.005076
Drums	AtomGS	26.452	24.320	0.004239	0.005301	0.007549
Drums	CoAdaptGS	19.025	13.911	0.167573	0.129150	0.164701
Drums	ConeGS	26.718	5.158	0.542299	0.002963	0.425044
Drums	erank-GS	26.719	25.559	0.000558	0.005011	0.004031
Drums	GaussianPro	27.240	23.185	0.006158	0.001807	0.008418
Drums	GES	26.820	25.591	0.000652	0.005756	0.004213
Drums	GHAP	2.564	17.784	0.759043	0.007344	0.595376
Drums	GOF	26.853	25.034	0.018549	0.005742	0.017664
Drums	GSLPM	23.476	22.838	0.001405	0.004524	0.004774
Drums	HoGS	26.777	25.398	0.001074	0.004296	0.004325
Drums	LapisGS	26.958	22.642	0.008020	0.002669	0.009978
Drums	LightGaussian	26.573	23.817	0.006538	0.004444	0.009115
Drums	MiniSplatting	26.510	7.541	0.290755	0.001737	0.231798
Drums	Mip-Splatting	27.163	24.673	0.003298	0.005188	0.006375
Drums	Octree-GS	26.929	19.643	0.017718	0.034031	0.025235
Drums	OP43DGS	26.987	25.329	0.001234	0.004097	0.004511
Drums	Opti3DGS	26.935	24.906	0.001798	0.005250	0.005231

Scene	Method	Black PSNR	White PSNR	Background excess	Foreground deficit	Matte MAE
Drums	PGSR	23.521	22.653	0.004495	0.002955	0.014948
Drums	PixelGS	27.086	23.209	0.005367	0.002327	0.007717
Drums	PUP 3D-GS	26.981	25.477	0.001085	0.004303	0.004366
Drums	ReactGS	27.054	21.548	0.009509	0.002757	0.010866
Drums	Scaffold-GS	27.816	22.001	0.010304	0.019671	0.015399
Drums	SOGS	26.805	24.750	0.002067	0.014616	0.007599
Drums	SteepGS	26.991	25.465	0.001082	0.004326	0.004359
Drums	TrimGS	20.542	1.912	0.822027	0.036916	0.653405
Drums	3DGS	23.781	22.924	0.005744	0.004474	0.015749
Ficus	2DGS	37.554	33.016	0.000207	0.009976	0.004728
Ficus	3DGS-MCMC	34.936	6.665	0.316630	0.018915	0.280549
Ficus	AbsGS	21.961	21.933	0.006210	0.001167	0.033984
Ficus	AnalyticSplatting	37.212	31.579	0.000222	0.013112	0.005713
Ficus	AtomGS	36.259	30.811	0.000435	0.011204	0.006129
Ficus	CoAdaptGS	18.139	15.433	0.098105	0.565467	0.154833
Ficus	ConeGS	35.843	20.858	0.055579	0.024304	0.055271
Ficus	erank-GS	36.243	30.850	0.000088	0.012973	0.006115
Ficus	GaussianPro	36.992	30.255	0.000676	0.008958	0.006553
Ficus	GES	36.573	30.762	0.000270	0.012150	0.006091
Ficus	GHAP	9.168	14.630	0.424315	0.015996	0.389222
Ficus	GOF	36.922	30.546	0.013237	0.011093	0.016588
Ficus	GSLPM	25.926	25.291	0.000659	0.011657	0.006518
Ficus	HoGS	36.709	30.545	0.000717	0.011469	0.006457
Ficus	LapisGS	36.576	30.460	0.000449	0.011281	0.006400
Ficus	LightGaussian	36.316	30.926	0.000585	0.012728	0.006489
Ficus	MiniSplatting	35.617	20.642	0.053941	0.012303	0.055430
Ficus	Mip-Splatting	37.379	31.400	0.000438	0.011781	0.005824
Ficus	Octree-GS	37.659	25.924	0.006808	0.022649	0.013901
Ficus	OP43DGS	36.694	30.468	0.000270	0.010988	0.006174
Ficus	Opti3DGS	36.644	30.497	0.000573	0.012404	0.006497
Ficus	PGSR	21.796	21.144	0.006614	0.000304	0.035484
Ficus	PixelGS	36.884	30.527	0.000343	0.010149	0.006091
Ficus	PUP 3D-GS	36.771	30.609	0.000266	0.010594	0.006037
Ficus	ReactGS	36.786	30.721	0.000252	0.010938	0.006021
Ficus	Scaffold-GS	37.290	27.882	0.005762	0.023024	0.012141
Ficus	SOGS	36.442	30.479	0.000558	0.021791	0.007280
Ficus	SteepGS	36.774	30.345	0.000701	0.010582	0.006411
Ficus	TrimGS	37.215	33.150	0.000054	0.006080	0.004266
Ficus	3DGS	21.971	22.098	0.006411	0.001698	0.033849
Hotdog	2DGS	38.389	28.785	0.001223	0.007300	0.003425
Hotdog	3DGS-MCMC	38.570	7.170	0.359323	0.003559	0.243866
Hotdog	AbsGS	32.048	28.126	0.001924	0.002236	0.004278
Hotdog	AnalyticSplatting	38.462	27.867	0.001598	0.007672	0.003886
Hotdog	AtomGS	37.162	21.581	0.011300	0.002618	0.009108
Hotdog	CoAdaptGS	22.522	18.459	0.137607	0.013317	0.100571
Hotdog	ConeGS	38.574	3.516	0.705154	0.000419	0.476009
Hotdog	erank-GS	37.850	27.462	0.002221	0.004109	0.003516
Hotdog	GaussianPro	38.929	24.553	0.005362	0.001429	0.005002
Hotdog	GES	37.945	29.229	0.000036	0.012088	0.003670
Hotdog	GHAP	4.674	17.078	0.636983	0.016506	0.438594
Hotdog	GOF	37.056	30.954	0.000057	0.011294	0.003788
Hotdog	GSLPM	27.688	25.714	0.001562	0.001675	0.002174
Hotdog	HoGS	38.064	30.342	0.000231	0.008993	0.002995
Hotdog	LapisGS	38.296	31.487	0.000468	0.005447	0.002269
Hotdog	LightGaussian	36.642	22.673	0.007886	0.002164	0.007553
Hotdog	MiniSplatting	38.062	8.849	0.208784	0.000355	0.145356

Scene	Method	Black PSNR	White PSNR	Background excess	Foreground deficit	Matte MAE
Hotdog	Mip-Splatting	38.815	29.346	0.001121	0.005593	0.002867
Hotdog	Octree-GS	38.385	5.566	0.431185	0.001363	0.297305
Hotdog	OP43DGS	38.424	29.656	0.000878	0.006330	0.002909
Hotdog	Opti3DGS	38.369	29.743	0.001177	0.003771	0.002467
Hotdog	PGSR	31.448	29.809	0.000730	0.008820	0.005076
Hotdog	PixelGS	38.658	24.170	0.005230	0.003661	0.005586
Hotdog	PUP 3D-GS	38.392	29.377	0.001036	0.006043	0.002962
Hotdog	ReactGS	38.529	18.799	0.021509	0.002410	0.017116
Hotdog	Scaffold-GS	38.437	6.248	0.377574	0.001185	0.261437
Hotdog	SOGS	38.100	31.308	0.000181	0.012628	0.004365
Hotdog	SteepGS	38.398	29.035	0.001159	0.005706	0.002978
Hotdog	TrimGS	37.084	11.316	0.141998	0.000787	0.094392
Hotdog	3DGS	32.078	25.418	0.002671	0.012868	0.008036
Lego	2DGS	38.286	24.416	0.001602	0.018683	0.007311
Lego	3DGS-MCMC	39.480	4.449	0.575646	0.000747	0.397077
Lego	AbsGS	28.332	27.261	0.003034	0.004423	0.007470
Lego	AnalyticSplatting	38.198	26.202	0.000457	0.019166	0.006837
Lego	AtomGS	35.139	31.348	0.000336	0.003870	0.002440
Lego	CoAdaptGS	23.505	13.679	0.201815	0.013953	0.146563
Lego	ConeGS	38.700	7.204	0.419939	0.001903	0.292195
Lego	erank-GS	37.916	33.258	0.000131	0.005027	0.002438
Lego	GaussianPro	39.288	21.031	0.012340	0.002602	0.010065
Lego	GES	37.034	26.102	0.000107	0.018334	0.006381
Lego	GHAP	4.137	14.305	0.793555	0.022612	0.553587
Lego	GOF	38.052	29.967	0.002158	0.013204	0.006101
Lego	GSLPM	29.448	27.264	0.001035	0.003745	0.002594
Lego	HoGS	37.691	22.462	0.006541	0.014562	0.009843
Lego	LapisGS	38.972	17.940	0.025104	0.003848	0.019736
Lego	LightGaussian	35.955	20.900	0.013749	0.006045	0.012237
Lego	MiniSplatting	38.586	12.846	0.091135	0.000797	0.065314
Lego	Mip-Splatting	39.246	31.250	0.000198	0.006806	0.002783
Lego	Octree-GS	37.994	8.642	0.230872	0.001606	0.161541
Lego	OP43DGS	38.345	27.762	0.000334	0.012153	0.004610
Lego	Opti3DGS	38.258	25.931	0.003012	0.009390	0.005842
Lego	PGSR	28.025	26.009	0.002321	0.017125	0.011017
Lego	PixelGS	39.263	20.002	0.015675	0.003112	0.012802
Lego	PUP 3D-GS	38.402	28.973	0.000169	0.010913	0.004166
Lego	ReactGS	39.007	14.736	0.055679	0.002426	0.040265
Lego	Scaffold-GS	37.677	8.772	0.218750	0.001613	0.154815
Lego	SOGS	37.421	26.595	0.004027	0.013928	0.008031
Lego	SteepGS	38.351	20.575	0.013856	0.010380	0.013962
Lego	TrimGS	38.460	33.210	0.000193	0.004007	0.002087
Lego	3DGS	28.269	25.921	0.002946	0.013156	0.009983
Materials	2DGS	31.739	29.115	0.000203	0.009543	0.003875
Materials	3DGS-MCMC	32.028	28.818	0.000125	0.015104	0.005433
Materials	AbsGS	30.105	25.476	0.003639	0.007332	0.008501
Materials	AnalyticSplatting	31.765	29.121	0.000200	0.010012	0.004011
Materials	AtomGS	30.918	28.085	0.000617	0.010484	0.004902
Materials	CoAdaptGS	22.219	12.669	0.213076	0.110174	0.187319
Materials	ConeGS	31.365	14.404	0.112892	0.010550	0.085002
Materials	erank-GS	31.253	28.614	0.000166	0.010042	0.004196
Materials	GaussianPro	32.033	29.855	0.000180	0.007169	0.003152
Materials	GES	31.300	28.678	0.000301	0.009995	0.004207
Materials	GHAP	20.415	17.572	0.158327	0.017833	0.122531
Materials	GOF	31.375	28.944	0.000309	0.009085	0.003774
Materials	GSLPM	21.504	20.904	0.000178	0.009449	0.003897

Scene	Method	Black PSNR	White PSNR	Background excess	Foreground deficit	Matte MAE
Materials	HoGS	31.365	28.948	0.000149	0.009230	0.003812
Materials	LapisGS	31.815	29.686	0.000216	0.007342	0.003168
Materials	LightGaussian	30.309	27.378	0.000951	0.011557	0.005539
Materials	MiniSplatting	30.863	23.526	0.044473	0.009151	0.035776
Materials	Mip-Splatting	32.081	29.747	0.000167	0.008401	0.003433
Materials	Octree-GS	32.845	28.672	0.002326	0.006825	0.004972
Materials	OP43DGS	31.691	29.215	0.000196	0.008942	0.003725
Materials	Opti3DGS	31.753	29.334	0.000156	0.008844	0.003661
Materials	PGSR	29.748	24.824	0.003260	0.012153	0.009455
Materials	PixelGS	32.017	29.628	0.000204	0.008075	0.003421
Materials	PUP 3D-GS	31.688	29.187	0.000177	0.009017	0.003722
Materials	ReactGS	31.952	29.425	0.000210	0.008688	0.003618
Materials	Scaffold-GS	32.615	28.668	0.002854	0.007292	0.005489
Materials	SOGS	31.253	28.320	0.000564	0.012799	0.005276
Materials	SteepGS	31.682	29.201	0.000194	0.008961	0.003722
Materials	TrimGS	31.648	28.436	0.000113	0.008096	0.003482
Materials	3DGS	30.142	25.893	0.003624	0.008539	0.008577
Mic	2DGS	36.921	32.979	0.000246	0.003204	0.001981
Mic	3DGS-MCMC	37.777	27.016	0.004588	0.026825	0.008725
Mic	AbsGS	30.574	25.470	0.001967	0.001108	0.008633
Mic	AnalyticSplatting	36.907	32.424	0.000359	0.005595	0.002469
Mic	AtomGS	35.674	32.140	0.000308	0.004277	0.002382
Mic	CoAdaptGS	23.230	15.260	0.106844	0.292245	0.129516
Mic	ConeGS	35.832	8.541	0.285696	0.015836	0.256633
Mic	erank-GS	36.548	32.415	0.000279	0.004254	0.002374
Mic	GaussianPro	37.528	29.635	0.001372	0.001255	0.003129
Mic	GES	36.447	32.543	0.000414	0.003774	0.002290
Mic	GHAP	2.043	20.763	0.732756	0.001806	0.652212
Mic	GOF	36.560	31.429	0.009686	0.006001	0.010696
Mic	GSLPM	27.719	26.843	0.000383	0.004333	0.002329
Mic	HoGS	36.296	32.669	0.000194	0.002743	0.001991
Mic	LapisGS	37.333	27.830	0.002885	0.001854	0.004469
Mic	LightGaussian	33.979	27.401	0.004128	0.002347	0.005927
Mic	MiniSplatting	35.943	6.870	0.322902	0.003189	0.290299
Mic	Mip-Splatting	37.469	30.404	0.001174	0.004193	0.003076
Mic	Octree-GS	37.925	16.702	0.039015	0.013310	0.037774
Mic	OP43DGS	37.245	32.876	0.000330	0.003145	0.002154
Mic	Opti3DGS	37.316	31.641	0.000366	0.008329	0.002891
Mic	PGSR	30.449	26.051	0.001976	0.001387	0.008106
Mic	PixelGS	37.516	31.820	0.000454	0.002598	0.002246
Mic	PUP 3D-GS	37.198	32.736	0.000321	0.003079	0.002141
Mic	ReactGS	37.589	26.218	0.005742	0.003369	0.007171
Mic	Scaffold-GS	37.451	29.277	0.003150	0.010090	0.005430
Mic	SOGS	36.435	28.807	0.002379	0.019565	0.006140
Mic	SteepGS	37.221	32.945	0.000254	0.002989	0.002059
Mic	TrimGS	31.140	3.090	0.573304	0.139064	0.530098
Mic	3DGS	30.600	25.641	0.002227	0.003482	0.008879
Ship	2DGS	32.986	28.788	0.000263	0.015325	0.008264
Ship	3DGS-MCMC	33.091	9.118	0.307504	0.003333	0.170708
Ship	AbsGS	29.537	26.433	0.002312	0.001774	0.007434
Ship	AnalyticSplatting	32.977	27.693	0.000361	0.020457	0.010607
Ship	AtomGS	32.467	29.270	0.001087	0.003549	0.003418
Ship	CoAdaptGS	22.665	14.885	0.153906	0.107258	0.130385
Ship	ConeGS	32.995	4.056	0.828191	0.000300	0.449438
Ship	erank-GS	32.725	30.717	0.000440	0.003708	0.003226
Ship	GaussianPro	33.372	29.636	0.000907	0.006348	0.004766

Scene	Method	Black PSNR	White PSNR	Background excess	Foreground deficit	Matte MAE
Ship	GES	32.670	26.815	0.000429	0.022608	0.011600
Ship	GHAP	4.266	12.744	0.751813	0.047014	0.430782
Ship	GOF	32.726	29.873	0.007937	0.008211	0.009053
Ship	GSLPM	22.440	21.786	0.001148	0.009482	0.006406
Ship	HoGS	32.672	28.846	0.000996	0.010397	0.006707
Ship	LapisGS	33.076	26.770	0.003477	0.005014	0.005623
Ship	LightGaussian	31.373	26.800	0.004386	0.006266	0.006988
Ship	MiniSplatting	32.507	8.824	0.267591	0.000873	0.149785
Ship	Mip-Splatting	33.368	28.149	0.002913	0.007130	0.006213
Ship	Octree-GS	33.288	8.544	0.281255	0.011131	0.165567
Ship	OP43DGS	33.056	28.104	0.000476	0.015969	0.008684
Ship	Opti3DGS	33.081	29.433	0.000938	0.008817	0.005848
Ship	PGSR	29.066	25.313	0.002271	0.010033	0.011436
Ship	PixelGS	33.222	28.736	0.000638	0.013050	0.007648
Ship	PUP 3D-GS	33.077	28.401	0.000289	0.015205	0.008299
Ship	ReactGS	33.042	28.125	0.001198	0.013180	0.008057
Ship	Scaffold-GS	33.132	7.803	0.328138	0.011271	0.192521
Ship	SOGS	32.624	24.298	0.001055	0.051806	0.026233
Ship	SteepGS	32.968	27.904	0.000710	0.015887	0.008760
Ship	TrimGS	25.522	2.793	0.785286	0.309308	0.586881
Ship	3DGS	29.498	22.779	0.003384	0.042382	0.026323

B.1. Cross-scene recurrence

For each scene, the highest-error group contains exactly eight methods, using descending background excess and method ID to resolve ties. The current boundary has no consequential ties. Spearman correlations compare complete thirty-method scene rankings; Kendall concordance summarizes their joint ordering.

Table 7. Descriptive recurrence statistics on the complete 30-by-8 matrix.

Statistic	Value
Median pairwise rank correlation	0.579978
Minimum pairwise rank correlation	0.331257
Maximum pairwise rank correlation	0.931924
Kendall concordance	0.654825
Raw matrix method sum-of-squares share	63.61%
Raw matrix scene sum-of-squares share	4.12%
Raw matrix nonadditive sum-of-squares share	32.26%

The last three rows are descriptive sums of squares in an unreplicated matrix. The nonadditive component combines departures from an additive method/scene description; repeated training is not available as a separate unit in this matrix.

C. Quality Sets and Sampling Support

Table 8. Every scene's near-best quality set, retaining its size, composition and error range.

Scene	States	Discovery	Extension	Median	Range	At least 0.05	At least 0.10
Chair	8	2	6	0.000071	0.000056–0.000131	0	0
Drums	2	0	2	0.008231	0.006158–0.010304	0	0
Ficus	6	3	3	0.000330	0.000054–0.006808	0	0
Hotdog	14	4	10	0.003414	0.000878–0.705154	4	4
Lego	6	2	4	0.020389	0.000198–0.575646	2	1
Materials	2	0	2	0.002590	0.002326–0.002854	0	0
Mic	8	2	6	0.003017	0.000454–0.039015	0	0
Ship	15	4	11	0.000938	0.000263–0.828191	4	4

Table 9. All sixty-one quality-set members. Selection uses only same-scene black-background PSNR.

Scene	Method	Set	Black PSNR	Background excess	Foreground deficit	Matte MAE
Chair	AnalyticSplatting	Discovery	36.443	0.000056	0.001184	0.000721
Chair	PUP 3D-GS	Extension	36.490	0.000057	0.000593	0.000670
Chair	OP43DGS	Extension	36.440	0.000062	0.000654	0.000693
Chair	SteepGS	Extension	36.471	0.000066	0.000621	0.000684
Chair	PixelGS	Extension	36.854	0.000077	0.000569	0.000681
Chair	Mip-Splatting	Discovery	36.919	0.000100	0.000699	0.000634
Chair	ReactGS	Extension	36.650	0.000121	0.000681	0.000747
Chair	LapisGS	Extension	36.556	0.000131	0.000563	0.000749
Drums	GaussianPro	Extension	27.240	0.006158	0.001807	0.008418
Drums	Scaffold-GS	Extension	27.816	0.010304	0.019671	0.015399
Ficus	TrimGS	Discovery	37.215	0.000054	0.006080	0.004266
Ficus	2DGS	Extension	37.554	0.000207	0.009976	0.004728
Ficus	AnalyticSplatting	Discovery	37.212	0.000222	0.013112	0.005713
Ficus	Mip-Splatting	Discovery	37.379	0.000438	0.011781	0.005824
Ficus	Scaffold-GS	Extension	37.290	0.005762	0.023024	0.012141
Ficus	Octree-GS	Extension	37.659	0.006808	0.022649	0.013901
Hotdog	OP43DGS	Extension	38.424	0.000878	0.006330	0.002909
Hotdog	PUP 3D-GS	Extension	38.392	0.001036	0.006043	0.002962
Hotdog	Mip-Splatting	Discovery	38.815	0.001121	0.005593	0.002867
Hotdog	SteepGS	Extension	38.398	0.001159	0.005706	0.002978
Hotdog	Opti3DGS	Extension	38.369	0.001177	0.003771	0.002467
Hotdog	2DGS	Extension	38.389	0.001223	0.007300	0.003425
Hotdog	AnalyticSplatting	Discovery	38.462	0.001598	0.007672	0.003886
Hotdog	PixelGS	Extension	38.658	0.005230	0.003661	0.005586
Hotdog	GaussianPro	Extension	38.929	0.005362	0.001429	0.005002
Hotdog	ReactGS	Extension	38.529	0.021509	0.002410	0.017116
Hotdog	3DGS-MCMC	Discovery	38.570	0.359323	0.003559	0.243866
Hotdog	Scaffold-GS	Extension	38.437	0.377574	0.001185	0.261437
Hotdog	Octree-GS	Extension	38.385	0.431185	0.001363	0.297305
Hotdog	ConeGS	Discovery	38.574	0.705154	0.000419	0.476009
Lego	Mip-Splatting	Discovery	39.246	0.000198	0.006806	0.002783
Lego	GaussianPro	Extension	39.288	0.012340	0.002602	0.010065
Lego	PixelGS	Extension	39.263	0.015675	0.003112	0.012802
Lego	LapisGS	Extension	38.972	0.025104	0.003848	0.019736
Lego	ReactGS	Extension	39.007	0.055679	0.002426	0.040265
Lego	3DGS-MCMC	Discovery	39.480	0.575646	0.000747	0.397077
Materials	Octree-GS	Extension	32.845	0.002326	0.006825	0.004972
Materials	Scaffold-GS	Extension	32.615	0.002854	0.007292	0.005489

Scene	Method	Set	Black PSNR	Background excess	Foreground deficit	Matte MAE
Mic	PixelGS	Extension	37.516	0.000454	0.002598	0.002246
Mic	Mip-Splatting	Discovery	37.469	0.001174	0.004193	0.003076
Mic	GaussianPro	Extension	37.528	0.001372	0.001255	0.003129
Mic	LapisGS	Extension	37.333	0.002885	0.001854	0.004469
Mic	Scaffold-GS	Extension	37.451	0.003150	0.010090	0.005430
Mic	3DGS-MCMC	Discovery	37.777	0.004588	0.026825	0.008725
Mic	ReactGS	Extension	37.589	0.005742	0.003369	0.007171
Mic	Octree-GS	Extension	37.925	0.039015	0.013310	0.037774
Ship	2DGS	Extension	32.986	0.000263	0.015325	0.008264
Ship	PUP 3D-GS	Extension	33.077	0.000289	0.015205	0.008299
Ship	AnalyticSplatting	Discovery	32.977	0.000361	0.020457	0.010607
Ship	OP43DGS	Extension	33.056	0.000476	0.015969	0.008684
Ship	PixelGS	Extension	33.222	0.000638	0.013050	0.007648
Ship	SteepGS	Extension	32.968	0.000710	0.015887	0.008760
Ship	GaussianPro	Extension	33.372	0.000907	0.006348	0.004766
Ship	Opti3DGS	Extension	33.081	0.000938	0.008817	0.005848
Ship	ReactGS	Extension	33.042	0.001198	0.013180	0.008057
Ship	Mip-Splatting	Discovery	33.368	0.002913	0.007130	0.006213
Ship	LapisGS	Extension	33.076	0.003477	0.005014	0.005623
Ship	Octree-GS	Extension	33.288	0.281255	0.011131	0.165567
Ship	3DGS-MCMC	Discovery	33.091	0.307504	0.003333	0.170708
Ship	Scaffold-GS	Extension	33.132	0.328138	0.011271	0.192521
Ship	ConeGS	Discovery	32.995	0.828191	0.000300	0.449438

Table 10. Complete versus near-best subsets, split by discovery/extension role. Each row contains observed members.

Subset	Set	States	Methods	Mean	Median	At least 0.05	At least 0.10
all	Discovery	64	8	0.131826	0.004257	23	20
all	Extension	176	22	0.048603	0.001065	23	21
Q_s	Discovery	17	5	0.164038	0.001174	5	5
Q_s	Extension	44	11	0.037749	0.001849	5	4

The discovery and extension groups have different selection roles. Their summaries describe these retained sets and are not estimates from independent random population samples. The sixteen/eighty-view comparison of the sixty-four discovery states has maximum absolute background-excess difference 0.010479 and matte-MAE difference 0.006870; each metric has two ranking changes. The primary task and quality membership consistently use sixteen views. The direct Hotdog LPIPS example is explicitly reported on eighty views.

D. Complete Fixed-State Effects and Area Controls

The first panel contains AnalyticSplatting, ConeGS and MiniSplatting on Hotdog and Drums. The extended panel contains PixelGS, ReactGS and GaussianPro on Lego and Mic. All 124 variants below use sixteen fixed cameras and their original checkpoints. Random seeds identify deletion draws, not new training. Positive background reduction and positive foreground increase retain the main-text sign conventions.

Table 11. All fixed-state variants. Direct PSNR pools clamped-float MSE before conversion to dB; direct LPIPS is the stored clamped-float output distance. GT PSNR loss compares each output with the original-background target.

State	Fraction	Selection	Removed	ΔB	ΔF	Direct PSNR	Direct LPIPS	GT PSNR loss
AnalyticSplatting/Drums	0.01%	Footprint	39	0.000004	0.000590	46.875	0.0018202	0.05177
AnalyticSplatting/Drums	0.01%	Scale	39	0.000145	0.000128	50.568	0.0007381	0.03259
AnalyticSplatting/Drums	0.01%	Random 0	39	0.000001	0.000002	71.046	0.0000026	0.00011
AnalyticSplatting/Drums	0.01%	Random 1	39	0.000000	0.000001	73.294	0.0000015	0.00011
AnalyticSplatting/Drums	0.01%	Random 2	39	0.000000	0.000004	70.483	0.0000027	0.00007
AnalyticSplatting/Drums	0.1%	Footprint	391	0.000029	0.006931	38.979	0.0062165	0.27721
AnalyticSplatting/Drums	0.1%	Scale	391	0.000218	0.002530	43.498	0.0030905	0.10090
AnalyticSplatting/Drums	0.1%	Random 0	391	0.000000	0.000018	62.268	0.0000169	0.00133
AnalyticSplatting/Drums	0.1%	Random 1	391	0.000000	0.000014	61.923	0.0000173	0.00178
AnalyticSplatting/Drums	0.1%	Random 2	391	0.000000	0.000013	59.884	0.0000200	0.00227
AnalyticSplatting/Drums	1%	Footprint	3912	0.000279	0.093414	25.897	0.0344583	3.66109
AnalyticSplatting/Drums	1%	Scale	3912	0.000351	0.025397	32.523	0.0133667	1.09750
AnalyticSplatting/Drums	1%	Random 0	3912	0.000006	0.000232	50.453	0.0002632	0.02252
AnalyticSplatting/Drums	1%	Random 1	3912	0.000005	0.000206	51.519	0.0002080	0.01569
AnalyticSplatting/Drums	1%	Random 2	3912	0.000007	0.000232	51.253	0.0002289	0.01606
AnalyticSplatting/Hotdog	0.01%	Footprint	17	0.000008	0.005066	57.397	0.0000779	0.05306
AnalyticSplatting/Hotdog	0.01%	Scale	17	0.000000	0.000274	61.469	0.0000485	0.02239
AnalyticSplatting/Hotdog	0.01%	Random 0	17	0.000018	0.000005	70.839	0.0000083	0.00200
AnalyticSplatting/Hotdog	0.01%	Random 1	17	0.000000	0.000000	77.580	0.0000008	0.00042
AnalyticSplatting/Hotdog	0.01%	Random 2	17	0.000000	0.000000	77.775	0.0000013	0.00046
AnalyticSplatting/Hotdog	0.1%	Footprint	172	0.000014	0.011451	43.966	0.0018864	1.08623
AnalyticSplatting/Hotdog	0.1%	Scale	172	0.000010	0.006011	52.839	0.0004993	0.15168
AnalyticSplatting/Hotdog	0.1%	Area control 0	163	0.000001	0.000601	54.247	0.0001907	0.09475
AnalyticSplatting/Hotdog	0.1%	Area control 1	163	0.000003	0.000656	54.992	0.0002005	0.10646
AnalyticSplatting/Hotdog	0.1%	Area control 2	163	0.000001	0.001289	52.840	0.0002934	0.16708
AnalyticSplatting/Hotdog	0.1%	Area target	163	0.000003	0.001892	52.947	0.0004920	0.14694
AnalyticSplatting/Hotdog	0.1%	Random 0	172	0.000000	0.000011	65.120	0.0000220	0.00845
AnalyticSplatting/Hotdog	0.1%	Random 1	172	0.000000	0.000002	65.475	0.0000177	0.00822
AnalyticSplatting/Hotdog	0.1%	Random 2	172	0.000000	0.000008	64.132	0.0000285	0.01223
AnalyticSplatting/Hotdog	1%	Footprint	1726	0.000032	0.034291	34.504	0.0174016	5.46242
AnalyticSplatting/Hotdog	1%	Scale	1726	0.000021	0.023796	38.362	0.0087172	3.09604
AnalyticSplatting/Hotdog	1%	Random 0	1726	0.000008	0.000096	54.939	0.0002117	0.09064
AnalyticSplatting/Hotdog	1%	Random 1	1726	0.000037	0.000240	55.087	0.0002121	0.08806
AnalyticSplatting/Hotdog	1%	Random 2	1726	0.000000	0.000203	54.441	0.0002220	0.11040
ConeGS/Drums	0.01%	Footprint	4	0.070036	0.000537	55.716	0.0000289	-0.00171
ConeGS/Drums	0.01%	Scale	4	0.029367	0.000152	69.529	0.0000013	-0.00075
ConeGS/Drums	0.01%	Random 0	4	0.000000	0.000001	73.897	0.0000027	0.00012
ConeGS/Drums	0.01%	Random 1	4	0.000000	0.000000	67.758	0.0000037	0.00058
ConeGS/Drums	0.01%	Random 2	4	0.000000	0.000001	73.148	0.0000020	-0.00007
ConeGS/Drums	0.1%	Footprint	49	0.406517	0.004642	47.811	0.0002240	0.01415
ConeGS/Drums	0.1%	Scale	49	0.336849	0.003231	49.825	0.0001858	0.00490
ConeGS/Drums	0.1%	Random 0	49	0.000013	0.000008	57.256	0.0000446	0.00183
ConeGS/Drums	0.1%	Random 1	49	0.000000	0.000009	58.328	0.0000385	0.00371
ConeGS/Drums	0.1%	Random 2	49	0.000000	0.000012	61.336	0.0000183	0.00119
ConeGS/Drums	1%	Footprint	492	0.536042	0.016324	33.113	0.0095461	0.89152
ConeGS/Drums	1%	Scale	492	0.538181	0.009586	40.694	0.0036586	0.12008

State	Fraction	Selection	Removed	ΔB	ΔF	Direct PSNR	Direct LPIPS	GT PSNR loss
ConeGS/Drums	1%	Random 0	492	0.000016	0.000102	48.158	0.0004149	0.03074
ConeGS/Drums	1%	Random 1	492	0.000003	0.000126	47.549	0.0004596	0.03706
ConeGS/Drums	1%	Random 2	492	0.000028	0.000111	47.535	0.0004729	0.03696
ConeGS/Hotdog	0.01%	Footprint	6	0.010479	0.000027	73.528	0.0000009	0.00127
ConeGS/Hotdog	0.01%	Scale	6	0.011014	0.000076	72.994	0.0000020	0.00086
ConeGS/Hotdog	0.01%	Random 0	6	0.000000	0.000000	78.076	0.0000008	0.00076
ConeGS/Hotdog	0.01%	Random 1	6	0.000000	0.000000	77.201	0.0000012	0.00037
ConeGS/Hotdog	0.01%	Random 2	6	0.000000	0.000000	86.391	0.0000016	0.00013
ConeGS/Hotdog	0.1%	Footprint	65	0.150912	0.000332	66.316	0.0000078	0.01274
ConeGS/Hotdog	0.1%	Scale	65	0.149728	0.000195	60.051	0.0000190	0.06239
ConeGS/Hotdog	0.1%	Random 0	65	0.000000	0.000003	64.802	0.0000226	0.01087
ConeGS/Hotdog	0.1%	Random 1	65	0.000000	0.000000	64.052	0.0000230	0.01050
ConeGS/Hotdog	0.1%	Random 2	65	0.000000	0.000002	65.225	0.0000230	0.01335
ConeGS/Hotdog	1%	Footprint	656	0.649984	0.014288	39.590	0.0070111	2.50610
ConeGS/Hotdog	1%	Scale	656	0.611412	0.011651	39.917	0.0036815	2.27559
ConeGS/Hotdog	1%	Random 0	656	0.000001	0.000107	53.913	0.0002327	0.12555
ConeGS/Hotdog	1%	Random 1	656	0.000008	0.000009	53.546	0.0002507	0.13089
ConeGS/Hotdog	1%	Random 2	656	0.000010	0.000023	53.710	0.0002612	0.13438
MiniSplatting/Drums	0.01%	Footprint	10	0.010350	0.000033	58.614	0.0000241	0.00169
MiniSplatting/Drums	0.01%	Scale	10	0.006258	0.000025	57.166	0.0000335	0.00093
MiniSplatting/Drums	0.01%	Random 0	10	0.000001	0.000001	67.187	0.0000019	-0.00027
MiniSplatting/Drums	0.01%	Random 1	10	0.000007	0.000000	72.052	0.0000018	-0.00001
MiniSplatting/Drums	0.01%	Random 2	10	0.000002	0.000000	65.361	0.0000042	0.00032
MiniSplatting/Drums	0.1%	Footprint	104	0.038656	0.000140	52.136	0.0001142	0.00307
MiniSplatting/Drums	0.1%	Scale	104	0.027816	0.000085	50.875	0.0001601	0.01305
MiniSplatting/Drums	0.1%	Random 0	104	0.000004	0.000005	59.963	0.0000339	0.00231
MiniSplatting/Drums	0.1%	Random 1	104	0.000022	0.000007	56.252	0.0000528	0.00378
MiniSplatting/Drums	0.1%	Random 2	104	0.000037	0.000006	60.104	0.0000256	0.00155
MiniSplatting/Drums	1%	Footprint	1048	0.133976	0.000778	45.221	0.0011207	0.05023
MiniSplatting/Drums	1%	Scale	1048	0.103205	0.000687	42.910	0.0014003	0.07634
MiniSplatting/Drums	1%	Random 0	1048	0.001442	0.000076	47.736	0.0004434	0.02970
MiniSplatting/Drums	1%	Random 1	1048	0.001729	0.000075	47.782	0.0004475	0.02822
MiniSplatting/Drums	1%	Random 2	1048	0.001223	0.000080	47.666	0.0005516	0.03258
MiniSplatting/Hotdog	0.01%	Footprint	10	0.007377	0.000014	70.004	0.0000126	0.00647
MiniSplatting/Hotdog	0.01%	Scale	10	0.000367	0.000015	70.899	0.0000065	0.00021
MiniSplatting/Hotdog	0.01%	Random 0	10	0.000000	0.000000	83.906	0.0000003	0.00009
MiniSplatting/Hotdog	0.01%	Random 1	10	0.000000	0.000000	75.290	0.0000034	0.00017
MiniSplatting/Hotdog	0.01%	Random 2	10	0.000001	0.000000	74.089	0.0000029	0.00127
MiniSplatting/Hotdog	0.1%	Footprint	107	0.023949	0.000129	60.302	0.0001234	0.03721
MiniSplatting/Hotdog	0.1%	Scale	107	0.013613	0.000142	58.362	0.0001681	0.04633
MiniSplatting/Hotdog	0.1%	Random 0	107	0.000011	0.000000	65.106	0.0000186	0.00862
MiniSplatting/Hotdog	0.1%	Random 1	107	0.000003	0.000000	64.369	0.0000391	0.01034
MiniSplatting/Hotdog	0.1%	Random 2	107	0.000009	0.000000	63.616	0.0000381	0.01504
MiniSplatting/Hotdog	1%	Footprint	1078	0.073919	0.006194	39.935	0.0055544	2.18446
MiniSplatting/Hotdog	1%	Scale	1078	0.052937	0.003826	43.143	0.0038881	1.21226
MiniSplatting/Hotdog	1%	Random 0	1078	0.000430	0.000006	53.865	0.0003197	0.11125
MiniSplatting/Hotdog	1%	Random 1	1078	0.000668	0.000007	53.919	0.0002831	0.09547
MiniSplatting/Hotdog	1%	Random 2	1078	0.002038	0.000003	54.521	0.0002738	0.09336
GaussianPro/Lego	0.1%	Footprint	1003	0.001373	0.001611	48.584	0.0009401	0.49451
GaussianPro/Lego	0.1%	Scale	1003	0.000630	0.000898	56.293	0.0003054	0.08659
GaussianPro/Lego	0.1%	Random 0	1003	0.000000	0.000003	64.354	0.0000111	0.01257
GaussianPro/Lego	0.1%	Random 1	1003	0.000000	0.000009	65.041	0.0000131	0.01103
GaussianPro/Lego	0.1%	Random 2	1003	0.000000	0.000012	64.190	0.0000152	0.01213
GaussianPro/Mic	0.1%	Footprint	1132	0.000234	0.000174	53.767	0.0001848	0.09542
GaussianPro/Mic	0.1%	Scale	1132	0.000301	0.000112	56.997	0.0000818	0.05057
GaussianPro/Mic	0.1%	Random 0	1132	0.000003	0.000003	67.925	0.0000037	0.00482

State	Fraction	Selection	Removed	ΔB	ΔF	Direct PSNR	Direct LPIPS	GT PSNR loss
GaussianPro/Mic	0.1%	Random 1	1132	0.000000	0.000006	66.374	0.0000058	0.00424
GaussianPro/Mic	0.1%	Random 2	1132	0.000000	0.000003	67.710	0.0000037	0.00458
PixelGS/Lego	0.1%	Footprint	980	0.003523	0.002084	47.740	0.0010309	0.60154
PixelGS/Lego	0.1%	Scale	980	0.001817	0.001376	54.918	0.0003333	0.12536
PixelGS/Lego	0.1%	Random 0	980	0.000000	0.000004	65.125	0.0000165	0.01098
PixelGS/Lego	0.1%	Random 1	980	0.000001	0.000003	66.856	0.0000095	0.00726
PixelGS/Lego	0.1%	Random 2	980	0.000000	0.000004	66.356	0.0000097	0.00740
PixelGS/Mic	0.1%	Footprint	456	0.000109	0.000240	53.026	0.0001474	0.09002
PixelGS/Mic	0.1%	Scale	456	0.000065	0.000175	54.563	0.0000926	0.06137
PixelGS/Mic	0.1%	Random 0	456	0.000000	0.000004	65.748	0.0000061	0.00628
PixelGS/Mic	0.1%	Random 1	456	0.000000	0.000003	65.235	0.0000079	0.00668
PixelGS/Mic	0.1%	Random 2	456	0.000000	0.000005	65.653	0.0000045	0.00865
ReactGS/Lego	0.1%	Footprint	491	0.012794	0.000872	51.583	0.0004214	0.24653
ReactGS/Lego	0.1%	Scale	491	0.001742	0.000206	61.588	0.0000532	0.02456
ReactGS/Lego	0.1%	Random 0	491	0.000001	0.000003	65.661	0.0000113	0.00861
ReactGS/Lego	0.1%	Random 1	491	0.000001	0.000007	65.402	0.0000130	0.00883
ReactGS/Lego	0.1%	Random 2	491	0.000001	0.000019	63.980	0.0000163	0.01426
ReactGS/Mic	0.1%	Footprint	239	0.002092	0.000273	54.028	0.0000841	0.04554
ReactGS/Mic	0.1%	Scale	239	0.000851	0.000070	64.668	0.0000121	0.00130
ReactGS/Mic	0.1%	Random 0	239	0.000000	0.000007	62.954	0.0000105	0.01306
ReactGS/Mic	0.1%	Random 1	239	0.000001	0.000016	65.045	0.0000056	0.00712
ReactGS/Mic	0.1%	Random 2	239	0.000000	0.000010	66.503	0.0000060	0.00620

Table 12. Spatial allocation at the fixed 0.1% fraction. Shares are per-view fractions of absolute alpha-change mass, then averaged. Positive matte improvement means lower error.

State	Selection	Background (%)	Foreground (%)	Boundary (%)	Matte improvement
AnalyticSplatting/Drums	Footprint	1.70	83.53	14.77	-0.001434
AnalyticSplatting/Drums	Scale	24.50	51.41	24.09	-0.000357
AnalyticSplatting/Hotdog	Footprint	0.26	99.68	0.06	-0.002959
AnalyticSplatting/Hotdog	Scale	0.30	99.65	0.05	-0.001407
ConeGS/Drums	Footprint	97.99	0.31	1.70	0.315125
ConeGS/Drums	Scale	98.14	0.26	1.60	0.260579
ConeGS/Hotdog	Footprint	99.51	0.13	0.36	0.098958
ConeGS/Hotdog	Scale	99.69	0.06	0.25	0.097936
GaussianPro/Lego	Footprint	66.12	33.03	0.85	0.000490
GaussianPro/Lego	Scale	65.22	31.87	2.92	0.000210
GaussianPro/Mic	Footprint	79.23	9.08	11.69	0.000206
GaussianPro/Mic	Scale	83.91	5.50	10.58	0.000272
MiniSplatting/Drums	Footprint	98.59	0.11	1.29	0.030028
MiniSplatting/Drums	Scale	98.79	0.09	1.13	0.021493
MiniSplatting/Hotdog	Footprint	99.56	0.40	0.05	0.016908
MiniSplatting/Hotdog	Scale	98.92	0.97	0.11	0.009330
PixelGS/Lego	Footprint	60.65	37.74	1.61	0.001967
PixelGS/Lego	Scale	66.28	29.37	4.35	0.000940
PixelGS/Mic	Footprint	62.80	23.77	13.43	0.000079
PixelGS/Mic	Scale	54.14	21.14	24.72	0.000055
ReactGS/Lego	Footprint	96.33	3.06	0.61	0.008491
ReactGS/Lego	Scale	92.72	6.15	1.13	0.001130
ReactGS/Mic	Footprint	93.12	1.63	5.25	0.001877
ReactGS/Mic	Scale	93.95	0.97	5.09	0.000761

D.1. Common area support

Table 13. The fixed common-support test on the original six states. A negative result is the measured failure to meet the prescribed matching support, rather than a completed equal-area control.

State	Positive-support count	Common count	Count (%)	Area (%)	Criterion met
AnalyticSplatting/Drums	377	346	91.78	73.30	No
AnalyticSplatting/Hotdog	167	163	97.60	87.17	Yes
ConeGS/Drums	49	26	53.06	38.97	No
ConeGS/Hotdog	59	43	72.88	57.17	No
MiniSplatting/Drums	99	91	91.92	72.72	No
MiniSplatting/Hotdog	106	90	84.91	69.43	No

The matching rule requires at least 80% common count and 80% common area and a total-area difference of at most 10%. Only AnalyticSplatting/Hotdog satisfies the support condition. Its target contains 163 primitives. The three evaluation-area ratios are 0.9367, 0.9316 and 0.9306. Footprint support sums individual primitive areas over the four scoring cameras and does not equal their union area or their composited contribution. The count-matched footprint comparisons therefore retain a difference in support dose.

E. Background-Conditioned Coverage Organization

The paired from-scratch core comprises AnalyticSplatting and ConeGS on Hotdog and Lego with seeds 0 and 1, each trained under B0, G05 and R for 30,000 actual Gaussian updates. B0 is black, G05 is RGB (0.5,0.5,0.5), and R independently samples the three channels from Uniform[0,1] at each update. Each method–scene–seed setting reuses its native initialization across the three backgrounds. ConeGS initialization includes its 20k NeRF stage, outside the Gaussian update count. AnalyticSplatting uses the existing L1/SSIM wrapper on the common 800-by-800 grid; the repository training path containing SGF is not used. The native filter and density-control paths remain method-specific.

Training reads train RGBA only: foreground RGB is premultiplied by its matte before applying the common image filter. The target and prediction use the same sampled background, with full-image RGB L1/SSIM and the declared native regularization, without an additional alpha or foreground-mask loss. Background RNG is separate from view order and native RNG. Endpoint scores use sixteen cameras and pooled MSE before conversion to PSNR; display-clamped image scores are shown below. Raw image scores remain in the accompanying endpoint tables.

Fixed-carrier refinement branches six original P5 checkpoints into three backgrounds and performs 3,000 actual updates per branch. Only opacity and enabled SH coefficients are optimized, using constant native initial learning rates: opacity 0.05, SH DC 0.0025 and other SH coefficients 0.000125. Position, scale, rotation, population, enabled SH degree, cameras and applicable filter geometry remain unchanged in all eighteen branches. All density-control and geometry-refresh operations are disabled. Geometry-preservation checks pass for every branch. P5 baselines and quality membership remain unchanged.

Trajectory evaluation uses the fixed four-view subset 0,20,40,60, including the final node. P7 uses 0,1k,3k,7k,15k,30k; P8 uses 0,100,300,1k,3k. Four-view endpoints are not replaced with sixteen-view summaries.

Table 14. Every paired-core from-scratch endpoint at 30k updates. Image scores use fixed black, gray and white targets.

Method	Scene	Seed	Bg.	B_{err}	F_{err}	Matte MAE	Black	Gray	White
AnalyticSplatting	Hotdog	0	B0	0.001053	0.009185	0.003764	38.492	33.417	28.493
AnalyticSplatting	Hotdog	0	G05	0.035570	0.000464	0.027277	22.197	38.827	22.199
AnalyticSplatting	Hotdog	0	R	1.25e-05	0.000304	0.000358	38.431	38.614	38.317
AnalyticSplatting	Hotdog	1	B0	0.001377	0.008135	0.003791	38.496	33.246	28.274
AnalyticSplatting	Hotdog	1	G05	0.026956	0.000470	0.021215	23.252	38.803	23.242
AnalyticSplatting	Hotdog	1	R	1.26e-05	0.000337	0.000365	38.384	38.558	38.263
AnalyticSplatting	Lego	0	B0	0.000299	0.017279	0.006175	38.202	32.344	27.095
AnalyticSplatting	Lego	0	G05	0.005947	0.004975	0.007089	29.437	37.855	29.353
AnalyticSplatting	Lego	0	R	6.27e-05	0.001040	0.000984	37.696	38.122	37.184
AnalyticSplatting	Lego	1	B0	0.000284	0.017098	0.006085	38.117	32.386	27.183
AnalyticSplatting	Lego	1	G05	0.006216	0.004470	0.007067	29.307	37.966	29.209
AnalyticSplatting	Lego	1	R	7.09e-05	0.001019	0.000989	37.795	38.238	37.224
ConeGS	Hotdog	0	B0	0.444952	0.000659	0.303234	38.621	11.743	5.728
ConeGS	Hotdog	0	G05	0.008226	0.004613	0.008636	28.743	38.699	28.763
ConeGS	Hotdog	0	R	1.55e-05	0.000698	0.000537	38.223	38.495	38.103
ConeGS	Hotdog	1	B0	0.503786	0.000375	0.348082	38.581	10.992	4.976
ConeGS	Hotdog	1	G05	0.017511	0.005736	0.016098	25.419	38.724	25.408
ConeGS	Hotdog	1	R	1.55e-05	0.000734	0.000547	38.246	38.504	38.108
ConeGS	Lego	0	B0	0.327757	0.002421	0.229713	38.753	14.408	8.397
ConeGS	Lego	0	G05	0.000777	0.012274	0.005532	33.948	39.100	33.795
ConeGS	Lego	0	R	4.04e-05	0.001142	0.001047	38.480	39.085	38.113
ConeGS	Lego	1	B0	0.310683	0.002513	0.217358	38.682	14.971	8.961
ConeGS	Lego	1	G05	0.001234	0.013465	0.006356	32.872	39.047	32.849
ConeGS	Lego	1	R	4.23e-05	0.001149	0.001061	38.388	38.999	38.028

Table 15. All same-scene, same-seed, same-background method comparisons. Gap is ConeGS minus AnalyticSplatting background excess; direct RGB compares their black-background outputs.

Scene	Seed	Bg.	Background gap	Direct RGB PSNR	Direct RGB LPIPS
Hotdog	0	B0	0.443898	45.509	0.002910

Scene	Seed	Bg.	Background gap	Direct RGB PSNR	Direct RGB LPIPS
Hotdog	0	G05	-0.027344	23.845	0.045495
Hotdog	0	R	2.98e-06	46.193	0.003040
Hotdog	1	B0	0.502409	45.238	0.002917
Hotdog	1	G05	-0.009445	27.878	0.029597
Hotdog	1	R	2.96e-06	46.136	0.003142
Lego	0	B0	0.327458	43.012	0.003140
Lego	0	G05	-0.005170	30.159	0.019453
Lego	0	R	-2.23e-05	42.407	0.003969
Lego	1	B0	0.310399	42.820	0.003433
Lego	1	G05	-0.004983	30.030	0.019105
Lego	1	R	-2.86e-05	42.651	0.003647

Table 16. Within-setting color and variation contrasts. Negative values indicate lower background excess. Every core setting is included.

Method	Scene	Seed	$B(\mathrm{G05}) - B(\mathrm{B0})$	$B(\mathrm{R}) - B(\mathrm{G05})$
AnalyticSplatting	Hotdog	0	0.034517	-0.035557
AnalyticSplatting	Hotdog	1	0.025579	-0.026943
AnalyticSplatting	Lego	0	0.005648	-0.005885
AnalyticSplatting	Lego	1	0.005932	-0.006145
ConeGS	Hotdog	0	-0.436726	-0.008210
ConeGS	Hotdog	1	-0.486275	-0.017495
ConeGS	Lego	0	-0.326980	-0.000737
ConeGS	Lego	1	-0.309449	-0.001191

Table 17. Every fixed-carrier endpoint at 3k updates. Positive background reduction improves coverage; positive foreground increase worsens undercoverage. Positive black PSNR cost denotes lower image quality than the original state.

State	Bg.	Initial B	Final B	ΔB	ΔF	PSNR cost	Direct LPIPS
AnalyticSplatting/Hotdog	B0	0.001598	0.001598	4.06e-07	-6.45e-05	-0.092	0.000324
AnalyticSplatting/Hotdog	G05	0.001598	2.35e-05	0.001575	-0.004291	+0.314	0.001116
AnalyticSplatting/Hotdog	R	0.001598	2.43e-05	0.001574	-0.005232	+0.144	0.000877
AnalyticSplatting/Lego	B0	0.000457	0.000458	-1.43e-06	-0.000101	-0.098	0.000200
AnalyticSplatting/Lego	G05	0.000457	5.95e-05	0.000397	-0.006452	+1.851	0.002811
AnalyticSplatting/Lego	R	0.000457	6.27e-05	0.000394	-0.009329	+0.672	0.001313
ConeGS/Hotdog	B0	0.705154	0.699560	0.005594	8.14e-06	-0.053	0.000590
ConeGS/Hotdog	G05	0.705154	0.007149	0.698005	0.004616	+9.090	0.014148
ConeGS/Hotdog	R	0.705154	0.000915	0.704239	0.001714	+2.102	0.004328
ConeGS/Lego	B0	0.419939	0.408222	0.011717	0.000115	-0.114	0.000439
ConeGS/Lego	G05	0.419939	0.000104	0.419835	0.002745	+0.643	0.001438
ConeGS/Lego	R	0.419939	0.000168	0.419771	0.000207	+0.150	0.000965
MiniSplatting/Hotdog	B0	0.208784	0.211111	-0.002328	1.41e-05	-0.172	0.000633
MiniSplatting/Hotdog	G05	0.208784	0.005819	0.202965	0.000921	+7.687	0.011839
MiniSplatting/Hotdog	R	0.208784	0.000421	0.208363	0.001462	+1.076	0.004176
MiniSplatting/Lego	B0	0.091135	0.094263	-0.003128	1.38e-05	-0.220	0.000424
MiniSplatting/Lego	G05	0.091135	0.000265	0.090870	0.000504	+0.943	0.002225
MiniSplatting/Lego	R	0.091135	5.28e-05	0.091082	0.000264	+0.089	0.000876

Table 18. Fixed-carrier task scores and unchanged nominal population. Black, gray and white are display-clamped PSNR (dB).

State	Bg.	Population	F_{err}	Matte MAE	Black	Gray	White
AnalyticSplatting/Hotdog	B0	172611	0.007607	0.003860	38.555	32.953	27.888
AnalyticSplatting/Hotdog	G05	172611	0.003381	0.001330	38.148	37.996	35.980
AnalyticSplatting/Hotdog	R	172611	0.002440	0.001037	38.318	38.049	36.377
AnalyticSplatting/Lego	B0	283455	0.019065	0.006809	38.296	31.603	26.229
AnalyticSplatting/Lego	G05	283455	0.012714	0.004674	36.347	35.179	30.159
AnalyticSplatting/Lego	R	283455	0.009837	0.003665	37.526	35.100	30.385
ConeGS/Hotdog	B0	65693	0.000427	0.472478	38.627	9.571	3.554
ConeGS/Hotdog	G05	65693	0.005035	0.007826	29.484	38.311	29.281
ConeGS/Hotdog	R	65693	0.002133	0.001794	36.472	37.910	35.110
ConeGS/Lego	B0	140881	0.002018	0.284497	38.815	13.151	7.137
ConeGS/Lego	G05	140881	0.004649	0.002352	38.057	38.913	36.936
ConeGS/Lego	R	140881	0.002110	0.001491	38.551	38.832	37.074
MiniSplatting/Hotdog	B0	107855	0.000369	0.146899	38.234	14.795	8.789
MiniSplatting/Hotdog	G05	107855	0.001276	0.005415	30.375	38.080	30.256
MiniSplatting/Hotdog	R	107855	0.001817	0.001288	36.986	37.781	36.002
MiniSplatting/Lego	B0	230748	0.000811	0.067426	38.805	18.655	12.662
MiniSplatting/Lego	G05	230748	0.001301	0.001418	37.643	38.999	36.926
MiniSplatting/Lego	R	230748	0.001061	0.001113	38.497	38.978	37.600

The optional extension adds one completed ScaffoldGS/Hotdog seed-0 black-background endpoint at 30k Gaussian updates: background excess 0.376526, foreground deficit 0.002244, and black/gray/white PSNR 38.295/12.304/6.291 dB. It is reported separately from the complete two-method paired core; the extension supplies a single black-background example.

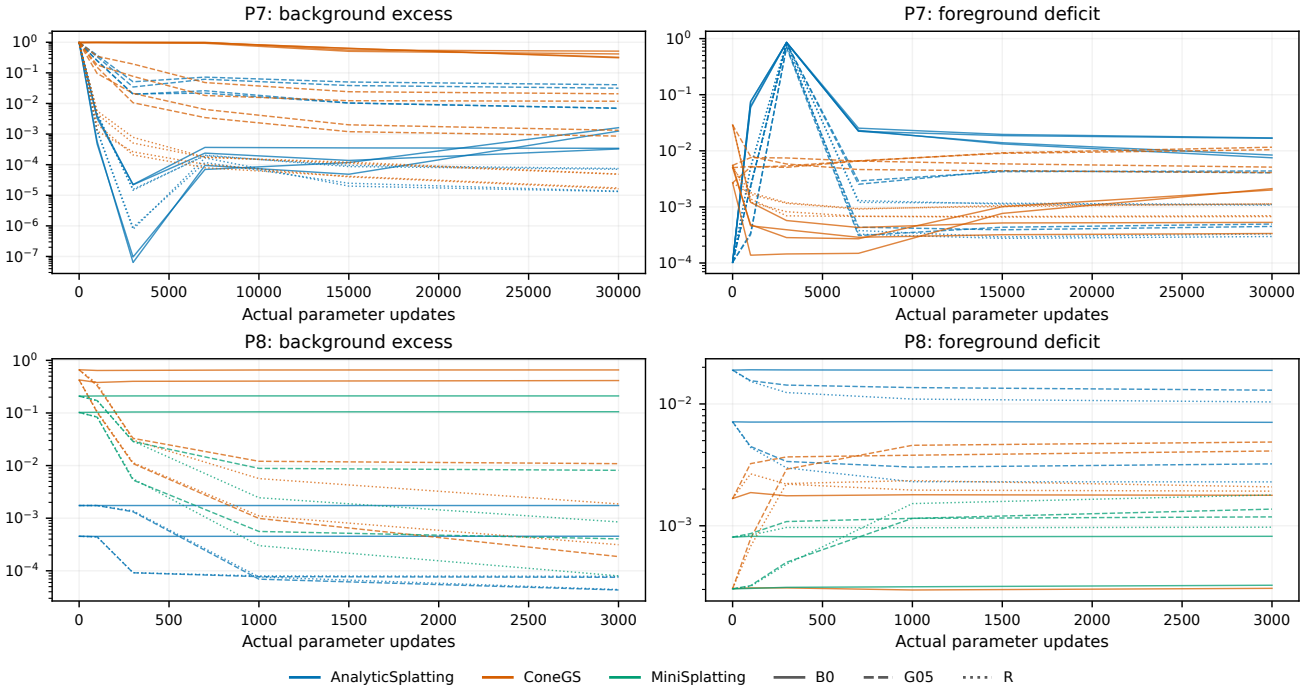


Figure 7. Four-view trajectories for all twenty-four paired-core P7 branches and eighteen P8 branches. Colors identify methods and line styles identify backgrounds. Each line retains one scene–seed–background branch. The tabulated trajectories also include the single ScaffoldGS extension.

Table 19. Complete four-view trajectories, including the single completed extension. The same four cameras define every node within each branch.

Study	Method	Scene	Seed	Bg.	Updates	B_{err}	F_{err}	Black PSNR
P7	AnalyticSplatting	Hotdog	0	B0	0	0.984907	0.000104	7.213
P7	AnalyticSplatting	Hotdog	0	B0	1000	0.000581	0.075208	27.874
P7	AnalyticSplatting	Hotdog	0	B0	3000	6.37e-08	0.850236	12.188
P7	AnalyticSplatting	Hotdog	0	B0	7000	9.46e-05	0.022465	33.987
P7	AnalyticSplatting	Hotdog	0	B0	15000	4.87e-05	0.013937	37.825
P7	AnalyticSplatting	Hotdog	0	B0	30000	0.001249	0.008409	39.130
P7	AnalyticSplatting	Hotdog	0	G05	0	0.984907	0.000104	7.213
P7	AnalyticSplatting	Hotdog	0	G05	1000	0.380321	0.000340	11.867
P7	AnalyticSplatting	Hotdog	0	G05	3000	0.051111	0.692968	13.072
P7	AnalyticSplatting	Hotdog	0	G05	7000	0.072674	0.000309	18.968
P7	AnalyticSplatting	Hotdog	0	G05	15000	0.050282	0.000435	20.542
P7	AnalyticSplatting	Hotdog	0	G05	30000	0.040909	0.000495	21.404
P7	AnalyticSplatting	Hotdog	0	R	0	0.984907	0.000104	7.213
P7	AnalyticSplatting	Hotdog	0	R	1000	0.001248	0.002343	27.675
P7	AnalyticSplatting	Hotdog	0	R	3000	7.75e-07	0.829022	12.192
P7	AnalyticSplatting	Hotdog	0	R	7000	0.000167	0.000326	34.690
P7	AnalyticSplatting	Hotdog	0	R	15000	1.99e-05	0.000275	37.747
P7	AnalyticSplatting	Hotdog	0	R	30000	1.32e-05	0.000297	38.982
P7	AnalyticSplatting	Hotdog	1	B0	0	0.985508	0.000103	7.211
P7	AnalyticSplatting	Hotdog	1	B0	1000	0.000477	0.075115	27.674
P7	AnalyticSplatting	Hotdog	1	B0	3000	9.61e-08	0.848196	12.198
P7	AnalyticSplatting	Hotdog	1	B0	7000	6.98e-05	0.023187	34.552
P7	AnalyticSplatting	Hotdog	1	B0	15000	0.000117	0.013271	38.091
P7	AnalyticSplatting	Hotdog	1	B0	30000	0.001616	0.007510	39.183
P7	AnalyticSplatting	Hotdog	1	G05	0	0.985508	0.000103	7.211
P7	AnalyticSplatting	Hotdog	1	G05	1000	0.338602	0.000318	12.586
P7	AnalyticSplatting	Hotdog	1	G05	3000	0.034494	0.717385	12.958
P7	AnalyticSplatting	Hotdog	1	G05	7000	0.061881	0.000435	19.715
P7	AnalyticSplatting	Hotdog	1	G05	15000	0.038553	0.000391	21.579
P7	AnalyticSplatting	Hotdog	1	G05	30000	0.031574	0.000449	22.430
P7	AnalyticSplatting	Hotdog	1	R	0	0.985508	0.000103	7.211
P7	AnalyticSplatting	Hotdog	1	R	1000	0.001330	0.002217	27.341
P7	AnalyticSplatting	Hotdog	1	R	3000	8.93e-07	0.829593	12.195
P7	AnalyticSplatting	Hotdog	1	R	7000	0.000118	0.000379	35.097
P7	AnalyticSplatting	Hotdog	1	R	15000	2.47e-05	0.000292	37.950
P7	AnalyticSplatting	Hotdog	1	R	30000	1.36e-05	0.000330	38.992
P7	AnalyticSplatting	Lego	0	B0	0	0.985298	0.000102	7.062
P7	AnalyticSplatting	Lego	0	B0	1000	0.003262	0.059183	24.686
P7	AnalyticSplatting	Lego	0	B0	3000	2.24e-05	0.860379	11.332
P7	AnalyticSplatting	Lego	0	B0	7000	0.000367	0.025352	34.180
P7	AnalyticSplatting	Lego	0	B0	15000	0.000353	0.019603	36.949
P7	AnalyticSplatting	Lego	0	B0	30000	0.000343	0.016983	38.184
P7	AnalyticSplatting	Lego	0	G05	0	0.985298	0.000102	7.062
P7	AnalyticSplatting	Lego	0	G05	1000	0.252327	0.002305	14.448
P7	AnalyticSplatting	Lego	0	G05	3000	0.020479	0.830969	11.388
P7	AnalyticSplatting	Lego	0	G05	7000	0.021762	0.002905	24.236
P7	AnalyticSplatting	Lego	0	G05	15000	0.010352	0.004250	27.196
P7	AnalyticSplatting	Lego	0	G05	30000	0.006911	0.004373	28.826
P7	AnalyticSplatting	Lego	0	R	0	0.985298	0.000102	7.062
P7	AnalyticSplatting	Lego	0	R	1000	0.004489	0.005283	24.633
P7	AnalyticSplatting	Lego	0	R	3000	1.46e-05	0.855407	11.321
P7	AnalyticSplatting	Lego	0	R	7000	0.000204	0.001207	34.182
P7	AnalyticSplatting	Lego	0	R	15000	8.65e-05	0.001164	36.493

Study	Method	Scene	Seed	Bg.	Updates	B_{err}	F_{err}	Black PSNR
P7	AnalyticSplatting	Lego	0	R	30000	7.07e-05	0.001102	37.629
P7	AnalyticSplatting	Lego	1	B0	0	0.985460	0.000102	7.061
P7	AnalyticSplatting	Lego	1	B0	1000	0.003831	0.062338	24.316
P7	AnalyticSplatting	Lego	1	B0	3000	2.20e-05	0.858335	11.346
P7	AnalyticSplatting	Lego	1	B0	7000	0.000238	0.022315	34.254
P7	AnalyticSplatting	Lego	1	B0	15000	0.000137	0.018658	37.111
P7	AnalyticSplatting	Lego	1	B0	30000	0.000326	0.016726	38.036
P7	AnalyticSplatting	Lego	1	G05	0	0.985460	0.000102	7.061
P7	AnalyticSplatting	Lego	1	G05	1000	0.249562	0.002349	14.535
P7	AnalyticSplatting	Lego	1	G05	3000	0.019968	0.831702	11.392
P7	AnalyticSplatting	Lego	1	G05	7000	0.026078	0.002534	23.495
P7	AnalyticSplatting	Lego	1	G05	15000	0.010347	0.004421	27.286
P7	AnalyticSplatting	Lego	1	G05	30000	0.007033	0.004051	28.769
P7	AnalyticSplatting	Lego	1	R	0	0.985460	0.000102	7.061
P7	AnalyticSplatting	Lego	1	R	1000	0.004806	0.004912	24.617
P7	AnalyticSplatting	Lego	1	R	3000	1.57e-05	0.860687	11.260
P7	AnalyticSplatting	Lego	1	R	7000	0.000195	0.001297	34.379
P7	AnalyticSplatting	Lego	1	R	15000	9.49e-05	0.001127	36.832
P7	AnalyticSplatting	Lego	1	R	30000	7.49e-05	0.001077	37.660
P7	ConeGS	Hotdog	0	B0	0	0.992716	0.028928	18.003
P7	ConeGS	Hotdog	0	B0	1000	0.974921	0.001208	29.914
P7	ConeGS	Hotdog	0	B0	3000	0.954755	0.000572	32.038
P7	ConeGS	Hotdog	0	B0	7000	0.924387	0.000430	35.185
P7	ConeGS	Hotdog	0	B0	15000	0.505881	0.000518	38.075
P7	ConeGS	Hotdog	0	B0	30000	0.415660	0.000530	39.187
P7	ConeGS	Hotdog	0	G05	0	0.992716	0.028928	18.003
P7	ConeGS	Hotdog	0	G05	1000	0.181034	0.007948	16.547
P7	ConeGS	Hotdog	0	G05	3000	0.076992	0.005825	21.469
P7	ConeGS	Hotdog	0	G05	7000	0.018231	0.004640	26.212
P7	ConeGS	Hotdog	0	G05	15000	0.012311	0.004394	26.692
P7	ConeGS	Hotdog	0	G05	30000	0.011837	0.004099	26.836
P7	ConeGS	Hotdog	0	R	0	0.992716	0.028928	18.003
P7	ConeGS	Hotdog	0	R	1000	0.001266	0.001387	30.101
P7	ConeGS	Hotdog	0	R	3000	0.000263	0.000694	32.382
P7	ConeGS	Hotdog	0	R	7000	0.000102	0.000678	35.528
P7	ConeGS	Hotdog	0	R	15000	3.99e-05	0.000660	37.738
P7	ConeGS	Hotdog	0	R	30000	1.63e-05	0.000673	38.794
P7	ConeGS	Hotdog	1	B0	0	0.998840	0.005494	18.241
P7	ConeGS	Hotdog	1	B0	1000	0.999543	0.000460	29.813
P7	ConeGS	Hotdog	1	B0	3000	0.999746	0.000392	31.886
P7	ConeGS	Hotdog	1	B0	7000	0.996860	0.000288	35.071
P7	ConeGS	Hotdog	1	B0	15000	0.555483	0.000319	38.235
P7	ConeGS	Hotdog	1	B0	30000	0.513070	0.000336	39.235
P7	ConeGS	Hotdog	1	G05	0	0.998840	0.005494	18.241
P7	ConeGS	Hotdog	1	G05	1000	0.347406	0.007565	13.315
P7	ConeGS	Hotdog	1	G05	3000	0.194440	0.007458	16.233
P7	ConeGS	Hotdog	1	G05	7000	0.048243	0.006579	21.364
P7	ConeGS	Hotdog	1	G05	15000	0.024196	0.005850	23.748
P7	ConeGS	Hotdog	1	G05	30000	0.020705	0.005122	24.433
P7	ConeGS	Hotdog	1	R	0	0.998840	0.005494	18.241
P7	ConeGS	Hotdog	1	R	1000	0.002099	0.001213	29.920
P7	ConeGS	Hotdog	1	R	3000	0.000208	0.000821	32.003
P7	ConeGS	Hotdog	1	R	7000	7.92e-05	0.000681	35.334
P7	ConeGS	Hotdog	1	R	15000	4.16e-05	0.000674	37.888
P7	ConeGS	Hotdog	1	R	30000	1.73e-05	0.000697	38.794
P7	ConeGS	Lego	0	B0	0	0.973632	0.002685	14.860

Study	Method	Scene	Seed	Bg.	Updates	B_{err}	F_{err}	Black PSNR
P7	ConeGS	Lego	0	B0	1000	0.982579	0.000138	16.911
P7	ConeGS	Lego	0	B0	3000	0.976788	0.000145	20.269
P7	ConeGS	Lego	0	B0	7000	0.964250	0.000149	23.553
P7	ConeGS	Lego	0	B0	15000	0.628896	0.000766	37.764
P7	ConeGS	Lego	0	B0	30000	0.324311	0.002119	38.872
P7	ConeGS	Lego	0	G05	0	0.973632	0.002685	14.860
P7	ConeGS	Lego	0	G05	1000	0.153163	0.005129	20.005
P7	ConeGS	Lego	0	G05	3000	0.010565	0.005373	28.225
P7	ConeGS	Lego	0	G05	7000	0.003456	0.006639	30.840
P7	ConeGS	Lego	0	G05	15000	0.001198	0.009015	33.363
P7	ConeGS	Lego	0	G05	30000	0.000859	0.010452	34.038
P7	ConeGS	Lego	0	R	0	0.973632	0.002685	14.860
P7	ConeGS	Lego	0	R	1000	0.005710	0.001668	31.713
P7	ConeGS	Lego	0	R	3000	0.000782	0.001156	33.842
P7	ConeGS	Lego	0	R	7000	0.000176	0.000915	35.731
P7	ConeGS	Lego	0	R	15000	0.000111	0.001061	37.669
P7	ConeGS	Lego	0	R	30000	4.86e-05	0.001127	38.480
P7	ConeGS	Lego	1	B0	0	0.965710	0.005016	13.785
P7	ConeGS	Lego	1	B0	1000	0.990598	0.000482	17.656
P7	ConeGS	Lego	1	B0	3000	0.983466	0.000284	20.978
P7	ConeGS	Lego	1	B0	7000	0.970560	0.000270	24.457
P7	ConeGS	Lego	1	B0	15000	0.636424	0.001009	38.031
P7	ConeGS	Lego	1	B0	30000	0.310404	0.002002	38.715
P7	ConeGS	Lego	1	G05	0	0.965710	0.005016	13.785
P7	ConeGS	Lego	1	G05	1000	0.095642	0.005242	21.145
P7	ConeGS	Lego	1	G05	3000	0.021394	0.005067	25.049
P7	ConeGS	Lego	1	G05	7000	0.006382	0.006559	28.945
P7	ConeGS	Lego	1	G05	15000	0.002006	0.009213	32.292
P7	ConeGS	Lego	1	G05	30000	0.001330	0.011600	32.948
P7	ConeGS	Lego	1	R	0	0.965710	0.005016	13.785
P7	ConeGS	Lego	1	R	1000	0.004522	0.001797	30.659
P7	ConeGS	Lego	1	R	3000	0.000508	0.001190	33.503
P7	ConeGS	Lego	1	R	7000	0.000174	0.000934	36.007
P7	ConeGS	Lego	1	R	15000	0.000121	0.001017	37.644
P7	ConeGS	Lego	1	R	30000	4.93e-05	0.001134	38.338
P7	ScaffoldGS	Hotdog	0	B0	0	0.984654	0.000104	7.186
P7	ScaffoldGS	Hotdog	0	B0	1000	0.118643	0.016738	29.067
P7	ScaffoldGS	Hotdog	0	B0	3000	0.113019	0.013200	33.039
P7	ScaffoldGS	Hotdog	0	B0	7000	0.189426	0.004092	36.591
P7	ScaffoldGS	Hotdog	0	B0	15000	0.277973	0.002159	37.852
P7	ScaffoldGS	Hotdog	0	B0	30000	0.362972	0.001598	38.856
P8	AnalyticSplatting	Hotdog	0	B0	0	0.001741	0.007132	39.007
P8	AnalyticSplatting	Hotdog	0	B0	100	0.001741	0.007099	39.035
P8	AnalyticSplatting	Hotdog	0	B0	300	0.001741	0.007106	39.230
P8	AnalyticSplatting	Hotdog	0	B0	1000	0.001740	0.007158	39.177
P8	AnalyticSplatting	Hotdog	0	B0	3000	0.001741	0.007065	39.146
P8	AnalyticSplatting	Hotdog	0	G05	0	0.001741	0.007132	39.007
P8	AnalyticSplatting	Hotdog	0	G05	100	0.001741	0.004476	38.731
P8	AnalyticSplatting	Hotdog	0	G05	300	0.001324	0.003372	38.785
P8	AnalyticSplatting	Hotdog	0	G05	1000	6.94e-05	0.003030	38.805
P8	AnalyticSplatting	Hotdog	0	G05	3000	4.32e-05	0.003222	38.725
P8	AnalyticSplatting	Hotdog	0	R	0	0.001741	0.007132	39.007
P8	AnalyticSplatting	Hotdog	0	R	100	0.001741	0.004394	38.779
P8	AnalyticSplatting	Hotdog	0	R	300	0.001377	0.002986	38.874
P8	AnalyticSplatting	Hotdog	0	R	1000	7.63e-05	0.002299	38.946
P8	AnalyticSplatting	Hotdog	0	R	3000	4.40e-05	0.002292	38.919

Study	Method	Scene	Seed	Bg.	Updates	B_{err}	F_{err}	Black PSNR
P8	AnalyticSplatting	Lego	0	B0	0	0.000453	0.018956	38.079
P8	AnalyticSplatting	Lego	0	B0	100	0.000451	0.019076	38.127
P8	AnalyticSplatting	Lego	0	B0	300	0.000452	0.019021	38.162
P8	AnalyticSplatting	Lego	0	B0	1000	0.000452	0.018965	38.209
P8	AnalyticSplatting	Lego	0	B0	3000	0.000454	0.018874	38.200
P8	AnalyticSplatting	Lego	0	G05	0	0.000453	0.018956	38.079
P8	AnalyticSplatting	Lego	0	G05	100	0.000442	0.015518	37.418
P8	AnalyticSplatting	Lego	0	G05	300	9.17e-05	0.014283	36.436
P8	AnalyticSplatting	Lego	0	G05	1000	7.74e-05	0.013625	36.183
P8	AnalyticSplatting	Lego	0	G05	3000	7.52e-05	0.012972	36.196
P8	AnalyticSplatting	Lego	0	R	0	0.000453	0.018956	38.079
P8	AnalyticSplatting	Lego	0	R	100	0.000442	0.015289	37.632
P8	AnalyticSplatting	Lego	0	R	300	9.21e-05	0.012404	37.300
P8	AnalyticSplatting	Lego	0	R	1000	8.00e-05	0.010964	37.303
P8	AnalyticSplatting	Lego	0	R	3000	7.86e-05	0.010388	37.272
P8	ConeGS	Hotdog	0	B0	0	0.662112	0.000305	39.007
P8	ConeGS	Hotdog	0	B0	100	0.642402	0.000309	39.086
P8	ConeGS	Hotdog	0	B0	300	0.645322	0.000309	39.312
P8	ConeGS	Hotdog	0	B0	1000	0.655959	0.000297	39.206
P8	ConeGS	Hotdog	0	B0	3000	0.657534	0.000306	39.173
P8	ConeGS	Hotdog	0	G05	0	0.662112	0.000305	39.007
P8	ConeGS	Hotdog	0	G05	100	0.330711	0.000783	30.811
P8	ConeGS	Hotdog	0	G05	300	0.032945	0.002914	26.364
P8	ConeGS	Hotdog	0	G05	1000	0.012070	0.004578	27.387
P8	ConeGS	Hotdog	0	G05	3000	0.010878	0.004868	27.419
P8	ConeGS	Hotdog	0	R	0	0.662112	0.000305	39.007
P8	ConeGS	Hotdog	0	R	100	0.360555	0.000678	28.617
P8	ConeGS	Hotdog	0	R	300	0.030448	0.002221	28.351
P8	ConeGS	Hotdog	0	R	1000	0.005661	0.002348	31.936
P8	ConeGS	Hotdog	0	R	3000	0.001868	0.002085	35.054
P8	ConeGS	Lego	0	B0	0	0.425033	0.001673	38.721
P8	ConeGS	Lego	0	B0	100	0.379910	0.001873	38.792
P8	ConeGS	Lego	0	B0	300	0.399575	0.001763	38.839
P8	ConeGS	Lego	0	B0	1000	0.403300	0.001797	38.925
P8	ConeGS	Lego	0	B0	3000	0.413197	0.001779	38.859
P8	ConeGS	Lego	0	G05	0	0.425033	0.001673	38.721
P8	ConeGS	Lego	0	G05	100	0.103107	0.003237	38.423
P8	ConeGS	Lego	0	G05	300	0.010903	0.003675	38.317
P8	ConeGS	Lego	0	G05	1000	0.000989	0.003794	38.251
P8	ConeGS	Lego	0	G05	3000	0.000186	0.004110	38.058
P8	ConeGS	Lego	0	R	0	0.425033	0.001673	38.721
P8	ConeGS	Lego	0	R	100	0.106411	0.002667	38.530
P8	ConeGS	Lego	0	R	300	0.011370	0.002197	38.544
P8	ConeGS	Lego	0	R	1000	0.001107	0.001962	38.637
P8	ConeGS	Lego	0	R	3000	0.000313	0.001924	38.507
P8	MiniSplatting	Hotdog	0	B0	0	0.208916	0.000304	38.318
P8	MiniSplatting	Hotdog	0	B0	100	0.210300	0.000306	38.326
P8	MiniSplatting	Hotdog	0	B0	300	0.210874	0.000312	38.617
P8	MiniSplatting	Hotdog	0	B0	1000	0.211129	0.000316	38.541
P8	MiniSplatting	Hotdog	0	B0	3000	0.210804	0.000325	38.585
P8	MiniSplatting	Hotdog	0	G05	0	0.208916	0.000304	38.318
P8	MiniSplatting	Hotdog	0	G05	100	0.172118	0.000322	38.175
P8	MiniSplatting	Hotdog	0	G05	300	0.028634	0.000502	31.383
P8	MiniSplatting	Hotdog	0	G05	1000	0.008846	0.001151	28.566
P8	MiniSplatting	Hotdog	0	G05	3000	0.008125	0.001370	28.666
P8	MiniSplatting	Hotdog	0	R	0	0.208916	0.000304	38.318

Study	Method	Scene	Seed	Bg.	Updates	B_{err}	F_{err}	Black PSNR
P8	MiniSplatting	Hotdog	0	R	100	0.172546	0.000318	38.195
P8	MiniSplatting	Hotdog	0	R	300	0.029037	0.000484	32.137
P8	MiniSplatting	Hotdog	0	R	1000	0.002450	0.001521	34.342
P8	MiniSplatting	Hotdog	0	R	3000	0.000848	0.001791	36.334
P8	MiniSplatting	Lego	0	B0	0	0.102393	0.000807	38.683
P8	MiniSplatting	Lego	0	B0	100	0.103306	0.000819	38.766
P8	MiniSplatting	Lego	0	B0	300	0.104338	0.000813	38.817
P8	MiniSplatting	Lego	0	B0	1000	0.105065	0.000813	38.882
P8	MiniSplatting	Lego	0	B0	3000	0.105516	0.000819	38.855
P8	MiniSplatting	Lego	0	G05	0	0.102393	0.000807	38.683
P8	MiniSplatting	Lego	0	G05	100	0.083438	0.000863	38.644
P8	MiniSplatting	Lego	0	G05	300	0.005299	0.001083	36.215
P8	MiniSplatting	Lego	0	G05	1000	0.000560	0.001151	37.269
P8	MiniSplatting	Lego	0	G05	3000	0.000405	0.001183	37.211
P8	MiniSplatting	Lego	0	R	0	0.102393	0.000807	38.683
P8	MiniSplatting	Lego	0	R	100	0.083843	0.000837	38.657
P8	MiniSplatting	Lego	0	R	300	0.005610	0.000968	36.225
P8	MiniSplatting	Lego	0	R	1000	0.000301	0.000963	38.458
P8	MiniSplatting	Lego	0	R	3000	8.01e-05	0.000973	38.485